

Model 5110

Dual Phase Lock-in Amplifier

Instruction Manual

222699-A-MNL-F

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BS EN50082-1 (1992):

IEC 801-2:1991

IEC 801-3:1994

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Warranty..... End of Manual

1.1 How to Use This Manual

This manual gives detailed instructions for setting up and operating the **SIGNAL RECOVERY** Model 5110 dual phase lock-in amplifier. It is split into the following chapters:-

Chapter 1 - Introduction

Provides an introduction to the manual, briefly describes what a lock-in amplifier is and the types of measurements it may be used for, and lists the major specifications of the model 5110.

Chapter 2 - Installation and Initial Checks

Describes how to install the instrument and gives a simple test procedure which may be used to check that the unit has arrived in full working order.

Chapter 3 - Technical Description

Provides an outline description of the design of the instrument and discusses the effect of the various controls. A good understanding of the design will enable the user to get the best possible performance from the unit.

Chapter 4 - Front and Rear Panels

Describes the connectors, controls and indicators which are found on the unit and which are referred to in the subsequent chapters.

Chapter 5 - Front Panel Operation

Describes the capabilities of the instrument when used as a manually operated unit, and shows how to operate it using the front panel controls.

Chapter 6 - Computer Operation

This chapter provides detailed information on operating the instrument from a computer over either the GPIB (IEEE-488) or RS232 interfaces. It includes information on how to establish communications, the functions available, the command syntax and a detailed command listing.

Appendix A

Gives the detailed specifications of the unit.

Appendix B

Details the pinouts of the multi-way connectors on the rear panel.

Appendix C

Lists three simple terminal programs which may be used as the basis for more complex user-written programs.

Appendix D

Shows the connection diagrams for suitable RS232 null-modem cables to couple the unit to an IBM-PC or 100% compatible computer.

Appendix E

Gives an alphabetical listing of the computer commands for easy reference.

New users are recommended to unpack the instrument and carry out the procedure in chapter 2 to check that it is working satisfactorily. They should then make themselves familiar with the information in chapters 3, 4 and 5, even if they intend that the unit will eventually be used under computer control. Only when they are fully conversant with operation from the front panel should they then turn to chapter 6 for information on how to use the instrument remotely. Once the structure of the computer commands is familiar, appendix E will prove convenient as it provides a complete alphabetical listing of these commands in a single easy-to-use section.

1.2 What is a Lock-in Amplifier?

In its most basic form the lock-in amplifier is an instrument with dual capability. It can recover signals in the presence of an overwhelming noise background or alternatively it can provide high resolution measurements of relatively clean signals over several orders of magnitude and frequency.

Modern instruments, such as the model 5110, offer far more than these two basic characteristics and it is this increased capability which has led to their acceptance in many fields of scientific research, such as optics, electrochemistry, materials science, fundamental physics and electrical engineering, as units which can provide the optimum solution to a wide range of measurement problems.

The model 5110 lock-in amplifier can function as a:-

- | | |
|---------------------------------|--------------------------|
| ■ AC Signal Recovery Instrument | ■ Oscillator |
| ■ Phase Meter | ■ Frequency Meter |
| ■ Spectrum Analyzer | ■ Noise Measurement Unit |

These characteristics, all available in a single compact unit, make it an invaluable addition to any laboratory.

1.3 Key Specifications and Benefits

The **SIGNAL RECOVERY** Model 5110 represents the ultimate in cost-effective analog lock-in amplifier technology, and offers:-

- Frequency range: 0.5 Hz to 100 kHz
- Voltage sensitivity: 100 nV to 1 V full-scale
- Current input mode sensitivities: 10 fA to 100 nA full-scale
10 fA to 1 μ A full-scale
- Line frequency rejection filter
- Dual phase demodulator with X-Y and R- θ outputs
- Fundamental (sine wave) or square-wave response modes
- Very low phase noise of < 0.005° rms.
- 3½-digit output readings
- Oscillator with variable amplitude and frequency
- Output time constants from <20 μ s to 300 s with 6 or 12 dB/octave roll-off
- Four external auxiliary ADC inputs
- Two external auxiliary DAC outputs
- Full range of auto-modes
- Standard GPIB (IEEE-488) and RS232 interfaces
- Bright LED indicators and seven-segment displays, and analog panel meter for control and display of instrument outputs

Installation & Initial Checks

Chapter 2

2.1 Installation

2.1.01 Introduction

Installation of the model 5110 in the laboratory or on the production line is very simple. Because of its low power consumption, the model 5110 does not incorporate forced-air ventilation. It can be operated on almost any laboratory bench or be rack mounted using the supplied accessory kit, at the user's convenience. With an ambient operating temperature range of 0°C to 35°C, it is highly tolerant to environmental variables, needing only to be protected from exposure to corrosive agents and liquids.

2.1.02 Rack Mounting

Two rack-mounting handles are supplied with the instrument which are fitted as follows:-

- 1) Turn the instrument off and disconnect the line power cord.
- 2) Using a suitable screwdriver, remove the two screws securing each of the two end-cheeks on each side of the front panel.
- 3) Fit the rack mounting handles using the same screws.
- 4) The four instrument feet may be removed if required, by removing the four screws securing them. However, if this is done, the bottom cover must be re-fixed in position using the shorter screws supplied with the rack-mounting handles.

CAUTION:- When the feet are removed, the maximum screw length that can be used is ¼" (6 mm). Use of longer screws could cause a short circuit that would damage the instrument and possibly create a dangerous shock hazard.

- 5) The instrument may now be mounted directly in a 19 inch rack using four securing bolts (not supplied)
- 6) Mechanical support should be provided for the rear of the instrument.

2.1.03 Inspection

Upon receipt the model 5110 Lock-in Amplifier should be inspected for shipping damage. If any is noted, **SIGNAL RECOVERY** should be notified immediately and a claim filed with the carrier. The shipping container should be saved for inspection by the carrier.

2.1.04 Line Cord Plug

A standard IEC 320 socket is mounted on the rear panel of the instrument and a suitable line cord is supplied.

2.1.05 Line Voltage Selection and Line Fuses

Before plugging in the line cord, ensure that the model 5110 is set to the voltage of the AC power supply to be used.

A detailed discussion of how to check and, if necessary, change the line voltage setting follows.

CAUTION: The model 5110 may be damaged if the line voltage is set for 110 V AC operation and it is turned on with 220 V AC applied to the power input connector.

The model 5110 can operate from any one of four different line voltage ranges, 90-110 V, 110-130 V, 200-240 V, and 220-260 V, at 50-60 Hz. The change from one range to another is made by repositioning a plug-in voltage selector circuit card located in the rear-panel power input assembly.

Observing the instrument from the rear, note the clear-plastic door immediately adjacent to the power cord connector (Figure 2-1). When the power cord is disconnected from the rear-panel connector, the plastic door is free to slide to the left, giving access to the fuse and to the voltage selector circuit card. The selector card is located at the lower edge of the fuse compartment. A number printed on the upper surface of the selector card is visible without removing the card. The number is somewhat obstructed by the fuse but can be read at the correct viewing angle. This number indicates the selected nominal line voltage. There are four numbers on the card, but only one is visible. In other words, the card can be inserted in one of four different positions, and a different number can be read in each. Table 2-1 indicates the actual line voltage range for each number. If the number showing is incorrect for the prevailing line voltage, the card will have to be repositioned, as follows.

The first step is to remove the fuse. When the lever labeled FUSE PULL is rotated out and towards the left, the fuse will lift so that it can be easily removed. At the front center of the circuit card is a small hole that serves as a convenient pry point. A small screwdriver or other tool can be used as an aid in removing the board. With the board removed, four numbers become visible: 100, 120, 220, and 240. Orient the board until the required number (Table 2-1) will be visible when the board is inserted. Then insert the board into its connector. The selected number should be the only one that shows. Ensure that the board is securely seated in its connector.

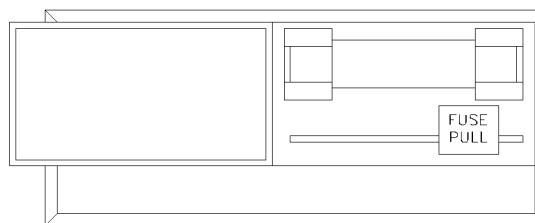


Figure 2-1. Power Input Assembly

Next check the fuse rating. For operation from a nominal line voltage of 120 V, use a slow-blow fuse rated at 1.0 A (voltage rating 250 V or higher). For operation from a nominal line voltage of 220 V, use a slow-blow fuse rated at 0.5 A (voltage rating of 250 V or higher). When the proper fuse has been installed, slide the plastic door back over the fuse compartment so that the power cord can be reconnected. Ensure that only fuses with the required current rating and of the specified type are used for replacement. The use of makeshift fuses and the short-circuiting of the fuse holder is prohibited and potentially dangerous.

VISIBLE #	VOLTAGE RANGE		
100	90	-	110 V
120	110	-	130 V
220	200	-	240 V
240	220	-	260 V

Table 2-1, Range Vs. Card Position

2.2 Initial Checks

2.2.01 Introduction

The following procedure checks the performance of the model 5110. In general, this procedure should be carried out after inspecting the instrument for obvious shipping damage.

NOTE: Any damage must be reported to the carrier and to **SIGNAL RECOVERY** immediately; take care to save the shipping container for inspection by the carrier.

Note that this procedure is intended to demonstrate that the instrument has arrived in good working order, not that it meets specifications. Each instrument receives a careful and thorough checkout before leaving the factory, and normally, if no shipping damage has occurred, will perform within the limits of the quoted specifications. If any problems are encountered in carrying out these checks, contact **SIGNAL RECOVERY** or the nearest authorized representative for assistance.

2.2.02 Procedure

- 1) Ensure that the Model 5110 is set to the line voltage of the power source (See Section 2.1.05 for details).
- 2) With the **POWER** switch set to off (push button at lower right-hand corner of front panel in the out position), plug the line cord into an appropriate source of power.
- 3) Turn on the power.
- 4) Check the instrument is in the LOCAL state (pushbutton in upper right-hand corner of the front panel). The **REMOTE** indicator directly above the pushbutton should NOT be lit.
- 5) Set the Model 5110 controls as follows.

SENSITIVITY Group

INPUT SELECT: **A** key depressed

FLOAT/GROUND: **FLOAT** key depressed

SENSITIVITY: Use the **▲** and **▼** keys until the legend **1 V** is lit

FILTERS Group

MODE: **BP** (BANDPASS)

TRACK: **TRACK** illuminated

DISPLAY 1 Group

DISPLAY 1: **OSC F** (internal oscillator frequency)

REFERENCE Group

REFERENCE MODE: **INT** (internal)

REFERENCE FREQUENCY MODE: **F** (**2F** extinguished)

DISPLAY 2 Group

TIME CONSTANT: Use the **▲** and **▼** keys until the legend **100 ms** is lit.

SLOPE: **6 dB** indicator lit

DYN RESERVE: **HI STAB**

OFFSET: Off (“**ON**” indicator extinguished)

EXPAND: Off (“**×10**” indicator extinguished)

DISPLAY: **% FS** and **X Y** lit

CONFIG/CNTL: Off (“**CONFIG**” and “**CNTL**” indicators extinguished)

- 6) Connect a cable between the **OSC OUT** connector and the **A** Input connector.
- 7) The **DISPLAY 1** display should indicate the **OSCILLATOR FREQUENCY** (display function selected in step 5). Use the **PARAMETER ▲** and **▼** keys to change the frequency to 1.000 kHz.

- 8) Press the **DISPLAY 1** select key once so that **OSC LVL** is lit. The display will indicate the amplitude of the oscillator output.
- 9) Use the **PARAMETER** ▲ and ▼ keys to set the output amplitude to 0.500 V.
- 10) Press the **DISPLAY 1** select key twice so that **DISP** is lit.
- 11) Press the red **AUTO** key and then the **MEAS** key (**SENSITIVITY** group ▼ key). This will initiate an auto-measure cycle. Automatically, the 5110 will:
 - a) Tune the band-pass signal channel filter to 1.000 kHz.
 - b) Set the full-scale sensitivity to 1 V range (already selected).
 - c) Adjust the reference phase for maximum output.

On completion of the auto-measure cycle (the red **AUTO** indicator will extinguish), both the analog meter and the **DISPLAY 1** group digital display should indicate a value close to 50% of full scale (input signal level of 0.500 V rms with selected sensitivity of 1 V rms). The **DISPLAY 2** indication will be small but unspecified (output of quadrature channel).

This completes the initial checks. If the indicated results were obtained, the user can be reasonably sure that the unit incurred no hidden damage in shipment and is in good working order.

3.1 Introduction

The model 5110 lock-in amplifier is a sophisticated instrument with many capabilities beyond those found in lower-cost instruments. This chapter discusses the various operating modes provided and then describes the design of the instrument by considering it as a series of functional blocks. In addition to describing how each block operates, each section also includes information on the effect of the various controls.

3.2 Operating Modes

3.2.01 Introduction

The model 5110 incorporates a number of different operating modes which are referred to in the following technical description, so in order to help the reader's understanding they are defined here.

3.2.02 Sine-Wave / Square-Wave Response

The fundamental purpose of a lock-in amplifier is to measure the amplitude of the component of the input voltage or current signal which is at the same frequency as that of the instrument's reference frequency. If the instrument gives an output only for signals at the reference frequency, then it is said to be *sine-wave* or *fundamental* responding.

In cases where the input signal takes the form of a square wave, for example if it is generated by detecting an optical signal passing through a rotating-blade light chopper, then it also contains components at odd harmonics of the fundamental frequency. Such harmonics contain useful information, and so if the instrument can be made to respond to these as well, a larger output will result. Instruments that respond in this way are known as *square-wave* responding units.

The model 5110 may operate in either mode, allowing the best mode to be chosen for all types of input signal.

3.2.03 2F Mode

Normally, a lock-in amplifier measures the applied signal at the reference frequency. However, in some applications such as Auger Spectroscopy and amplifier characterization, it is useful to be able to make measurements at twice the reference frequency. The model 5110 may be set to a "2F" mode which allows this, with the only restriction in performance being that the 2F signal cannot be greater than the upper frequency limit of the instrument, which is 100 kHz.

3.2.04 Internal / External Reference Mode

In the internal reference mode, the instrument's reference channel is driven by a signal derived from its internal oscillator. The oscillator signal is used to drive the

experiment, and the instrument measures the response of the experiment to this stimulation.

In the external reference mode, the experiment includes some device, for example an optical chopper, which generates a reference frequency which is applied to the lock-in amplifier's external reference input. The instrument's reference channel "locks" to this signal and uses it to measure the applied input signal.

3.3 Principles of Operation

3.3.01 Block Diagram

The model 5110 utilizes low-noise analog signal processing circuitry and a 68000 series microprocessor to achieve its specifications. A block diagram of the instrument is shown in figure 3-1, and the sections that follow describe how each functional block operates and the effect it has on the instrument's performance.

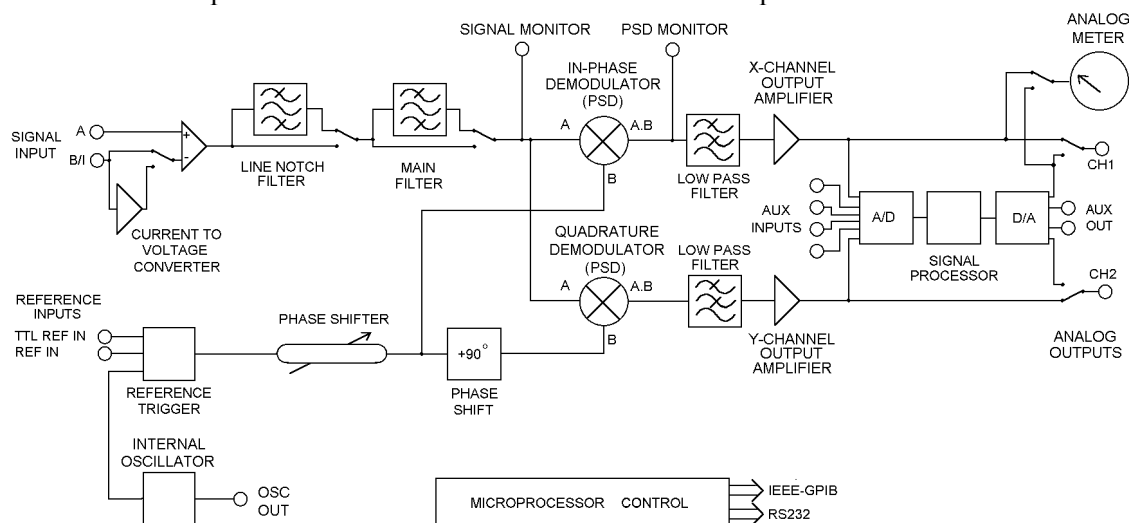


Figure 3-1, Model 5110 - Block Diagram

3.3.02 Signal-Channel Inputs

The signal input amplifier can be configured for either single-ended or differential voltage mode operation, or single-ended current mode operation. In voltage mode a choice of AC or DC coupling is available. In current mode two conversion gains may be selected to give optimum matching to the signal input. In both modes the input connector shells may be either floated via a 1 k Ω resistor or grounded to the instrument's chassis ground. These various features are discussed in the following paragraphs.

Input Connector Selection, A / A - B

When set to the **A** mode, the lock-in amplifier measures the voltage between the center and the shell of the **A** input BNC connector, whereas when set to the **A-B** mode it measures the difference in voltage between the center pins of the **A** and **B/I VIR GND** input BNC connectors.

The latter, differential, mode is often used to eliminate ground loops, although it is worth noting that at very low signal levels it may be possible to make a substantial reduction in unwanted offsets by using this mode, with a short-circuit terminator on the **B/I VIR GND** connector, rather than by simply using the **A** input mode.

The specification defined as the Common Mode Rejection Ratio, CMRR, defines how well the instrument rejects common mode signals applied to the **A** and **B/I VIR GND** inputs when operating in differential input mode. It is usually given in decibels. Hence a specification of > 100 dB implies that a common mode signal (i.e. a signal simultaneously applied to both **A** and **B/I VIR GND** inputs) of 1 V will give rise to less than 10 μ V of signal out of the input amplifier.

Input Connector Shell Ground / Float

The input connector shells may be connected either directly to the instrument's chassis ground or they can be floated by being connected via a 1 k Ω resistor. When in the float mode, the presence of this resistor substantially reduces the problems which often occur in low-level lock-in amplifier measurements due to ground loops.

Input Signal Selection, V / I

Although the voltage mode input is most commonly used, a current to voltage converter may be switched into use to provide current mode input capability, in which case the signal is connected to the **B/I VIR GND** connector. High impedance sources (> 100 k Ω) are inherently current sources and typically need to be measured with a low impedance current mode input. Even when dealing with a voltage source in series with a high impedance, the use of the current mode input may provide advantages in terms of improved bandwidth and immunity from the effects of cable capacitance.

The converter may be set to **10⁶ V/A** or **10⁸ V/A** conversion settings, but it should be noted that the -3 dB bandwidth when used in current input mode is much lower than that in voltage input mode. Better performance can be achieved using a separate current preamplifier, such as the **SIGNAL RECOVERY** model 5182.

3.3.03 Line Frequency Rejection Filter

Following the signal input amplifier, there is an option to pass the signal through a line frequency rejection filter, which is designed to give greater than 34 dB of attenuation within a $\pm 1\%$ range of the power line frequencies of 50 Hz or 60 Hz and their second harmonics at 100 Hz or 120 Hz.

The line frequency filter is supplied set to match the line frequency of the country of destination, but it will be appreciated that if a unit is moved from a 50 Hz area to a 60 Hz area then the filter will need to be adjusted. This can be done via one of the computer interfaces using the **LINE** command.

3.3.04 Main Filter

The main filter has two functions. First, it prevents large interfering signals, which would otherwise cause overload and non-linear operation, from reaching the phase sensitive detectors (PSDs). Consequently, it increases the instrument's *dynamic*

reserve (discussed later in section 3.3.15). Second, in conjunction with the special Walsh mode of operation of the demodulators, it causes the instrument to behave as if it were a sine-wave responding unit. The filter prevents signals at the harmonics of the reference frequency higher than the seventh from reaching the demodulator, whilst the Walsh demodulator rejects the third and fifth harmonics.

The main signal channel filter has three possible settings, as follows:-

Flat

In the flat mode of operation, the main filter is switched out of use and the demodulators are set to square-wave mode, giving an overall square-wave instrument response. Odd harmonics at the reference frequency of an applied square wave contribute to the output by a ratio $1/n$, where “ n ” is the order of the harmonic. Consequently, if the input signal is a square wave, the output will be about 23% higher with the flat mode selected than with the other settings, since in the latter cases only the fundamental frequency component of the square wave contributes to the output.

Low-Pass

In the low-pass setting, the filter is configured to give maximum rejection of frequencies above the set filter frequency, giving the best possible harmonic rejection. This setting is usually used when the interference is concentrated at frequencies higher than the reference frequency.

Bandpass

In the bandpass setting, the filter still rejects frequencies above the set filter frequency, but with less attenuation than in the low-pass case, but in addition it also rejects signals at lower frequencies. This setting is usually used when the interference is concentrated at frequencies lower than the reference frequency.

The filter is composed of two stacked second-order sections giving an overall fourth-order response and its frequency can be tuned manually or can automatically be set to track the reference frequency, over the full frequency range of the instrument. The tracking mode is useful in swept-frequency experiments, but if the reference frequency is stable then better performance will be achieved by setting the filter frequency control to the manual mode.

3.3.05 Signal Monitor Output

The analog signal following the input amplifiers, line frequency rejection filter and main signal channel filter is available at the front-panel **SIG MON** BNC connector. Because of the operation of the algorithm that distributes the overall gain between the signal channel and the output filters, the gain between the **A** or **A-B** input connectors and this connector depends on a number of factors. These include the full-scale sensitivity range, the main signal channel filter setting, the setting of the dynamic reserve control and the setting of the output filter time-constants. This should be taken into account when using the **SIG MON** output to monitor correct lock-in amplifier operation or to implement a stand-alone tuned amplifier.

The signal monitor output is unbuffered, and has an output impedance of 450 Ω . As

such, it should not be connected to loads of less than 15 k Ω . Internal noise and switching spikes may be present. The higher the reserve setting, the greater their amplitude will be relative to that of the signal.

The output of the signal channel is fed to the signal inputs of the in-phase and quadrature phase sensitive detectors (PSDs). Before discussing these in detail and the output stages that follow, the reference channel that provides the other input to the demodulators will be described.

3.3.06 Reference Channel

The reference channel in the instrument is responsible for implementing the reference trigger/phase-locked loop and phase shifter circuits and providing switching waveforms to the phase sensitive detectors.

There are two reference modes, as follows:

External Reference Mode

In external reference mode the reference source is applied to either a general purpose input, designed to accept virtually any periodic waveform with a 50:50 mark-space ratio and of suitable amplitude, or to a TTL-logic level input. Wherever possible, the TTL reference input should be used since it gives better phase accuracy.

Internal Reference Mode

With internal reference operation the output of the instrument's internal oscillator is internally coupled to the reference channel.

Regardless of the reference mode used, the reference circuits measure the actual reference frequency using the instrument's internal crystal oscillator clock. Above 2 kHz, the unit counts the number of reference cycles in a period of 0.5 s giving a measurement time of 0.5 s. Below 2 kHz it counts the number of cycles of a 4 MHz clock in one reference period and so the measurement time is inversely proportional to the frequency, with the longest time being 2 s at the lowest possible frequency of 0.5 Hz.

Although this measurement time is not important in most experiments, in those cases where the reference frequency is being stepped to a series of values, perhaps under computer control, it should be allowed for in determining the time needed to obtain an accurate measure of frequency following a change.

The time needed for the reference circuits to acquire lock following a change in frequency, the *lock acquisition time*, is nominally 2 reference periods plus 1 s. The red front-panel reference **UNLK** indicator lights whenever the reference circuit is not locked.

Within the reference circuits, the full operating frequency range of the instrument is split into five bands, with changeover points at 2 Hz, 20 Hz, 200 Hz, 2 kHz and 20 kHz. Normally, these need not concern the user, but if the reference frequency is stepped through one of them then the instrument will briefly lose lock, requiring that a

time equal to the lock acquisition time plus the output filter settling time be allowed for to obtain a steady output reading.

When the instrument is switched to 2F mode the reference circuits run at twice the applied reference frequency, with an upper limit for the latter of 50 kHz.

3.3.07 Internal Oscillator

In common with many other lock-in amplifiers the model 5110 incorporates an internal oscillator which may be used to drive an experiment. It is of the function-generator type, giving a sinusoidal output with good stability of amplitude and frequency and immediate settling after changes in frequency. The oscillator also generates a squarewave which, in the internal reference mode, is internally connected to the reference channel. Note that this internal squarewave shows a phase shift at higher frequencies, and, if the best phase accuracy is required, the external reference mode should be used by coupling the **OSC OUT** connector to the front-panel **REF IN** connector and using a suitable oscillator amplitude. However, in many applications the phase shift is not a problem because its effect can easily be counteracted by the use of the reference phase shifter.

The main disadvantage of this type of oscillator is its relatively high harmonic distortion of up to 0.5 percent. In most lock-in applications this is not an important matter because of the frequency-selective nature of the measurement, but, when the signal-channel filters are not in use, the distortion may lead to errors of a few tenths of one percent in amplitude measurements and phase errors of one or two tenths of a degree.

The output impedance of the oscillator is nominally 450 Ω , giving a useful 10:1 attenuation when loaded with 50 Ω . Note that when using the output with a typical cable and a high-impedance load, phase shifts of a few degrees may occur at higher frequencies.

The frequency of the oscillator is set from the front panel or by means of a computer command to any value between 0.5 Hz and 100 kHz. Although the setting resolution is better than 0.1 percent, the accuracy depends on component values and factory settings and is specified as ± 1 percent. On the other hand, the reference frequency meter as discussed above is based on a digital counter and a quartz crystal. The accuracy of this meter is better than 0.1 percent, and it can be used to monitor the frequency of the oscillator.

The internal amplitude of the oscillator is 5 V rms. This is available at the front-panel **OSC OUT** connector by the use of the computer command OA 5000, but this is not the normal method of operation. Normally the **OSC OUT** connection is taken from a two-stage attenuator consisting of a multiplying digital-to-analog converter (MDAC) followed by a switchable divide-by-ten stage. The control is by means of an integer in the range 0 to 2000 covering nominally 0 to 2000 mV rms, which can be set from the front panel or by means of the OA command.

Although this method of amplitude control is very convenient in practice, the user should bear in mind that the resolution with which the oscillator amplitude can be set

depends on whether the divide-by-ten stage is in use, so that it is 1 mV for voltages below 200 mV, but 8 mV at higher levels. Also the size of the steps is not exactly uniform and in particular, there may be a slight discontinuity at the transition between 200 mV and 201 mV, where the divide-by-ten stage is switched in or out and the MDAC code changes between 200 and 20.

3.3.08 Reference Monitor

A rear-panel **TTL REF OUT** BNC connector provides a TTL waveform at the applied reference frequency, either internal or external, and can be used to monitor correct reference channel operation.

3.3.09 Phase Shifter

The reference channel also implements the reference phase shifter, allowing the phase of the reference inputs to the demodulators be adjusted to the required value. The reference phase is defined as the phase of the reference input to the in-phase demodulator with respect to a sinusoidal reference input applied to the **REF IN** socket.

This means that when the reference phase is zero and the signal input to the demodulator is a full-scale sinusoid in phase with the reference input sinusoid, the output of the in-phase, or X channel, demodulator is a full-scale positive value and the quadrature, or Y channel, demodulator output is zero.

The circuits connected to the **REF IN** socket actually detect a positive-going crossing of the mean value of the applied reference voltage. Therefore when the reference input is not sinusoidal, its effective phase is the phase of a sinusoid with positive-going zero crossing at the same point in time, and accordingly the reference phase is defined with respect to this waveform. Similarly, the effective phase of a reference input to the **TTL REF IN** socket is that of a sinusoid with positive-going zero crossing at the same point in time.

In basic lock-in amplifier applications the purpose of the experiment is to measure the amplitude of a signal which is of fixed frequency and whose phase with respect to the reference input does not vary. This is a *scalar* measurement, for example the measurement of the response of a system to a chopped optical beam. Many other lock-in amplifier applications are of the *signed scalar* type. In these, the purpose of the experiment is to measure the amplitude and sign of a signal which is of fixed frequency and whose phase with respect to the reference input does not vary apart from reversals of phase corresponding to changes in the sign of the signal. A well known example of this situation is the case of a resistive bridge, one arm of which contains the sample to be measured. Other examples occur in derivative spectroscopy, where a small modulation is applied to the angle of the grating (in optical spectroscopy) or to the applied magnetic field (in magnetic resonance spectroscopy). Double beam spectroscopy is a further common example.

In these signed scalar measurements the phase shifter must be set, after removal of any zero errors, to maximize the X channel or the Y channel demodulator outputs. This is the only method that will give correct operation as the output signal passes through zero, and is also the best method to be used in an unsigned scalar measurement where

any significant amount of noise is present.

The reference phase shifter setting resolution can be set to 0.005° , 0.025° or 0.1° , although the front-panel control is limited to a resolution of 0.1° . In addition a **+90°** key allows the phase shift to be incremented in 90° steps.

The output from the phase shifter is directly connected to the reference input of the in-phase demodulator, and, via a further fixed 90° phase shifter to the reference input of the quadrature demodulator.

3.3.10 Demodulators - Introduction

Since the 5110 is a dual-phase instrument, there are two phase sensitive detectors (PSDs) or demodulators, one driven in quadrature to the other.

There are various ways of implementing such elements in an analog instrument, but in the 5110 they are based on FET switches giving excellent performance at reasonable cost. In conjunction with the main signal channel filter (described earlier in section 3.3.04) they operate in either the squarewave or Walsh modes, so that the instrument has an overall response which is either squarewave or sine-wave.

3.3.11 Demodulators - Squarewave Mode

Squarewave demodulator operation is automatically selected when the main signal channel filter is set to Flat mode. When selected, the outputs of the reference channel are squarewaves at the fundamental frequency of the reference input voltage, with a phase difference of 90 degrees between the channels. The multiplying elements consist of reversing switches under the control of the squarewaves generated by the reference channel. In practice each reversing switch consists of a two-way switch which causes the signal path to be connected alternately to the output of the signal channel and the output of an inverting amplifier (i.e. an amplifier with transfer function equal to -1) the input of which is connected to the output of the signal channel. Functionally, this arrangement acts as an analog multiplier which multiplies the output of the signal channel with a demodulation function consisting of a reference-derived squarewave having the two values 1 and -1. The Fourier analysis of this waveform consists of the fundamental and all odd harmonics, the amplitude of the n th harmonic being proportional to $1/n$. (Note that the fundamental is regarded as the first harmonic). It can be shown that the squarewave demodulator gives a steady-state output resulting from any Fourier component of the signal channel output which is at the fundamental frequency of the reference input voltage or any of its odd harmonics, with the response being inversely proportional to the harmonic number. Also, interfering signal voltages at frequencies close to the odd-harmonic frequencies can cause unwanted beat frequencies at the output.

3.3.12 Demodulators - Walsh Mode

The squarewave demodulator is simple to implement and (because the multiplication is performed only by switches) is capable of excellent performance at low cost. However, in the majority of experimental situations the odd-harmonic responses of the squarewave demodulator are undesirable because they implement “windows” in the

frequency domain response through which interfering voltages at or near the odd harmonics of the reference frequency can cause output errors, in the form of static offsets or of low-frequency beats. In principle, the problem could be solved by the use of a fundamental-only demodulator. In this case, the outputs of the reference channel are sinusoids at the fundamental frequency of the reference input voltage, with a phase difference of 90 degrees between the channels. The multipliers themselves are simply standard analog multipliers (i.e. multiplying elements which accept continuously variable voltages at both inputs) with one input connected to an output of the reference channel and the other to the output of the signal channel. Therefore the overall demodulator response is sinusoidal and there are no responses to reference harmonics other than the first.

In practice, there are severe problems both in implementing an analog multiplier of sufficiently good performance (in terms of frequency response, offsets and linearity) and also in constructing a reference channel to generate the required reference-derived sinusoids with sufficient purity over the frequency range required in general-purpose lock-in amplifiers. Therefore, the most common way of removing harmonic responses has been with the aid of signal-channel filters which reduce the amplitude of the offending components of the signal before they reach the demodulator. A general-purpose lock-in amplifier is normally equipped with signal-channel filters for quite a different reason, which is to increase the dynamic range by attenuating high-level interfering signals which would otherwise force the signal-channel gain to be reduced in order to prevent overloading. The preferred Q -factor of such filters, particularly for those under microprocessor control, is not more than 2 in order to avoid excessive phase errors, but this does not give sufficient rejection of reference harmonics.

What is required is a demodulator which like the squarewave demodulator is implemented with switches, but which has better harmonic rejection. An obvious solution is to use more switches, and a set of scaling resistors, to give an overall response which is intermediate between squarewave and a sinusoidal. The required coefficients can be calculated by the application of Walsh analysis, which is the counterpart of Fourier analysis in the domain of switching functions. An 8-step Walsh synthesis of a sinusoid yields a function having zero levels of third, fifth, eleventh, thirteenth, etc., harmonic, the amplitudes of the non-zero harmonics being the same as for a squarewave.

The demodulators in the model 5110 operate in this Walsh mode whenever the main signal channel filter is set to other than the Flat setting. When the filter is set to Bandpass mode, it has a fourth order response with a Q -factor of about 2, which in conjunction with the demodulator gives essentially an overall fundamental, or sine-wave, instrument response. The greatest harmonic response is at seven times the reference frequency, where the attenuation is about 1400 (63 dB). Where harmonic response is particularly critical a further factor of 50 (34 dB) is available by switching the main filter to the Low-Pass mode.

3.3.13 Output Filters - Operation

The time variation of the output of the lock-in amplifier should represent the time variation of the magnitude and phase of the required input signal. The function of the output filters is to reduce the level of spurious (i.e. unwanted, non-information

bearing) time variations, commonly referred to as output noise which may be random or deterministic in nature. One inevitable source of deterministic output noise results from the shape of the signal waveform. For example, the familiar rectified sinusoid waveform with mean value of unity varies from zero to $(\pi/2)$ twice in each cycle. A second-order filter with time constant equal to the period of the sinusoid will reduce the time variation to about ± 0.02 , and a filter with three times this time constant will leave a time variation less than ± 0.002 . Where the residual fluctuation is less than 1 least significant bit of the output analog to digital converter (ADC), a noise-free input sinusoid gives rise to a noise-free digital output. However, where additive noise (random or deterministic) is present, the output time constant will normally be increased to a value which reflects a compromise between output noise and response time. Note that for rejection of deterministic noise the second-order output filter is greatly superior to the first-order filter for equivalent response time.

If the noise process is random with an approximately flat spectrum (white noise) then the root-mean-square value of the output noise is inversely proportional to the square root of the time-constant of the output filters.

The output filters used in the model 5110 implement first-order or second-order low-pass functions, specified by the value of the time constant. For time-constant values of 30 ms or less, the output filters are conventional analog stages implemented with resistors, capacitors and operational amplifiers. Each filter has two identical first-order low-pass sections, one of which can be switched out, allow it to be used optionally as a first-order filter or as a second-order filter with two equal time constants.

When the time constant is equal to 100 ms or greater, digital filtering techniques carried out in the output processor are used in conjunction with the analog filters, but the same type of function is implemented. In this case, the analog X and Y outputs are obtained from digital-to-analog converters.

The time constant of the output filters can be set to values between 1 ms and 300 s in a 1-3-10 sequence by the front panel keys or by a computer command. A MIN setting is also available at which the time constant is less than 100 μ s, typically 20 μ s. The order of the output filters is controlled by the Slope key, which may seem somewhat strange, and a few words of explanation may be helpful.

In traditional audio terminology, a first-order low-pass filter is described as having "a slope of 6 dB per octave" because in the high-frequency limit its gain is inversely proportional to frequency (6 dB is approximately a factor of 2 in amplitude and an octave is a factor of 2 in frequency); similarly a second-order low-pass filter is described as having "a slope of 12 dB per octave". These terms have become part of the accepted terminology relating to lock-in amplifier output filters and are used in the 5110.

Accordingly the front-panel control which selects the order of the output filters is labeled **SLOPE** and the options are **6 dB** and **12 dB**. The 6 dB/octave option is not satisfactory for most purposes because it does not give good rejection of non-random interfering signals which can cause aliasing problems with the instrument's analog-to-digital converters. However, the 6 dB/octave filter finds use where the lock-in amplifier is incorporated in a feedback control loop, and in some situations where the

form of the time-domain response is critical. The user is recommended always to use 12 dB/octave unless there is some definite reason for not doing so.

3.3.14 Output Filters - Output Offset & Expand

Even when the input to the signal channel is nominally zero, the X-channel and Y-channel demodulator outputs may show non-zero values. This phenomenon is called zero error and is usually caused by unwanted coupling or crosstalk between the signal channel and the reference channel, either in the external connections or possibly (in the most sensitive ranges and at the highest reference frequencies) within the instrument itself. Zero errors may also be caused by the effects of changing ambient temperature on the demodulator hardware although these are usually negligible in the case of modern designs such as the model 5110. The magnitudes of any zero errors are usually dependent on the reference frequency.

The problem of crosstalk in the external circuit (i.e. the hardware which is generating the signal and the reference) is illustrated by the following numerical example. Suppose that the signal source impedance is 1 k Ω , and the reference is 1 V rms at 100 kHz. Then a stray capacitance of only 0.001 pF between the reference source and the lock-in amplifier's signal input connector would give an offset corresponding to 600 nV at the input - more than full scale on the three most sensitive ranges!

Unless they are large enough to cause overload, zero errors do not give rise to any malfunction in the demodulator, simply acting as additive errors which can be measured under zero-signal conditions and subsequently subtracted from the X and Y channel outputs. Consequently, in the model 5110 the output filters include an output offset facility allowing offsets of up to ± 100 % of full-scale to be applied to the X-channel and Y-channel outputs. The offset level can be set manually or via the Auto-Offset function.

Since the magnitude and signal phase outputs are derived by a non-linear operation on the X-channel and Y-channel outputs, it is essential that when using these outputs any X and Y zero errors are first reduced to a small fraction of full scale.

The X-channel output filter also includes an output expand facility, which allows a $\times 10$ expansion to be applied to the X output only, thereby increasing the maximum full-scale sensitivity to 10 nV.

3.3.15 Dynamic Reserve and Output Stability

The 5110 lock-in amplifier includes gain in three main areas, these being the signal channel ahead of the filters, between the filters and the demodulators, and in the output channels. The first two can be AC coupled, but the output stages have to be DC coupled. DC coupled amplifiers potentially exhibit DC drift with time and temperature, the effect of which increases as the gain increases. Hence it might at first sight appear that it would be best to use high values of AC gain and minimal DC gain. However, if this were done then the instrument would have very low *dynamic reserve*.

The dynamic reserve of a lock-in amplifier is defined as the ratio by which input noise voltages may exceed the full-scale sensitivity value without causing overload. It is

usually expressed in decibels:

$$\text{DR}(\text{in dB}) = 20 \times \log(\text{DR}(\text{as a ratio}))$$

Note that a noise voltage may be random or may be periodic at some fixed frequency; in the latter case it is often referred to as an interfering signal.

Obviously the dynamic reserve value is limited by the full-scale sensitivity setting: for example, a lock-in amplifier could well show a reserve value of 1000 (60 dB) at a full-scale sensitivity setting of 1 mV (implying the capability of handling 1 V noise) but would not be able to maintain this value of reserve at a full-scale sensitivity setting of 1 V because this would require the capability of handling 1000 V noise.

There are two mechanisms by which dynamic reserve is achieved, demodulator reserve and filter reserve, discussed in the following paragraphs.

Demodulator Reserve

The demodulator reserve is affected by the total signal channel (AC) gain. If this is high, then the demodulator will overload at lower levels of interfering signal than would be the case with lower gains. Consequently the dynamic reserve will be lower. Hence the maximum reserve is achieved at low levels of signal channel gain.

However at high sensitivity settings, lower values of AC gain require that higher values of DC gain are used to maintain the overall gain. Consequently the instrument exhibits poorer output stability with time and temperature.

Filter Reserve

One of the functions of the signal-channel filters is to provide dynamic reserve by reducing the amplitude of interfering signals. (The other function is to reject reference harmonics as discussed in sections 3.3.04 and 3.3.12). The success of this operation depends on the frequency of the interfering signal being well outside the pass-band of the filter. This is very often the case, because the pass-band occupies only a tiny fraction of the total frequency range; however, when a serious interfering signal does occur in the pass-band, it is sometimes possible to change the frequency of the measurement, for example, by changing the chopper speed in the case of an optical measurement. Note that the action of the filter does not improve the dynamic reserve unless there is amplification in the signal channel following the filter. If such gain is not present, an interfering signal of peak value equal to the overload level at the input of the filter would not cause overload whether attenuated by the filter or not. It follows from these considerations that the greatest improvement in the reserve that the filter can provide is equal to the value of the voltage gain in the signal channel following the signal-channel filter. This value is called the filter reserve.

In many experimental situations where the signal-channel filter is used for reserve improvement, the amount of available filter reserve is not the limiting factor.

In practice, the tunable fourth-order active filter shows substantially higher levels of

random noise than a simple amplifying stage, and if this is to be made insignificant compared with the preamplifier noise then there must be sufficient voltage gain in the signal channel in front of the filter.

Therefore the distribution of the total instrument gain involves two tradeoffs:

- 1) In the signal channel (AC) gain, gain in front of the filter improves the noise while gain after the filter improves the filter reserve.
- 2) Increasing the signal channel (AC) gain and reducing the output (DC) gain improves the output stability but decreases the dynamic reserve.

The model 5110 allows the user to control the distribution of these gains via the dynamic reserve control, which has three settings: high reserve, normal and high stability. The dynamic reserve and output stability available at each setting is shown in table 3-1

Setting	Dynamic Reserve		Stability
	Filter On	Filter Off	
High Reserve	up to 105 dB	60 dB	500 ppm/°C
Normal	up to 85 dB	40 dB	50 ppm/°C
High Stability	up to 65 dB	20 dB	5 ppm/°C

Table 3-1, Dynamic Reserve and Output Stability

The high reserve setting may cause increased demodulator noise and signal-channel noise, and reduced output stability. In most typical measurement situations a dynamic reserve of 20 dB is adequate and the use of the high stability setting is recommended. The normal setting, giving 40 dB demodulator reserve for full-scale sensitivity 10 mV and below, will handle very high noise levels (noise voltage 100 times the signal voltage), so that the high reserve mode should be used only in exceptional cases.

At lower values of full-scale sensitivity the instrument provides substantial filter reserve even in the high stability setting and the first choice for normal experimental situations should be high stability in conjunction with the bandpass or low-pass filter.

3.3.16 Main Analog to Digital Converter

The main analog to digital converter (ADC) in the 5110 converts the analog signals at the outputs of the X-channel and Y-channel output filters and at the rear-panel **ADC 1** to **ADC 4** auxiliary inputs to allow them to be displayed on the front-panel LED displays and read via the computer interfaces. In addition, the digitized signals can also be subjected to further processing by the output processor, as is required for example for magnitude and phase calculations and for noise measurements.

The ADC can be triggered in a number of ways, as follows:-

Reference Synchronous Trigger

In this setting, the ADC is triggered synchronously with the reference frequency. The maximum conversion rate is nominally once every 5 ms (200 Hz), so that at

low reference frequencies there will be one trigger per reference period. At higher reference frequencies there will be one trigger per some multiple of the reference period. Each trigger results in the conversion of X- and Y-channel outputs and one of the four auxiliary **ADC** inputs. Thus any given auxiliary **ADC** input is converted once every four trigger cycles.

External Trigger Mode

In this setting the **ADC** is triggered once on each positive going TTL transition detected at rear-panel **ADC EXT TRIG IN** connector. If an applied trigger occurs before the converter has finished processing the previous trigger, it is ignored. On any given trigger, the X- and Y-channel outputs and all four of the auxiliary **ADC** inputs are converted.

Asynchronous Trigger Mode

In this setting the **ADC** is triggered asynchronously at the preset system sample rate, which is adjustable over the range 100 Hz to 200 Hz by means of a front panel control or remote computer command. On any given trigger, the X- and Y-channel outputs and one of the four auxiliary **ADC** inputs are converted. Thus any given auxiliary **ADC** input is converted once every four trigger cycles (i.e. at a quarter of the system sample rate).

Ratio Trigger Mode

In this setting, triggering is the same as for the Reference Synchronous mode, but only the X- and Y- channel outputs and the auxiliary **ADC 1** input are converted. The maximum conversion rate is nominally once every 5 ms (200 Hz), so that at low reference frequencies there will be one trigger per reference period. At higher reference frequencies there will be one trigger per some multiple of the reference period.

The advantage of this mode over the reference synchronous trigger mode is that the auxiliary **ADC 1** input is converted at the same rate as the X- and Y-channel outputs, giving better results when the instrument is set to the ratio display mode.

NOTE: In the ratio trigger mode, the ADC 2, 3 and 4 inputs are never converted.

“8F” Trigger Mode

In this setting, at reference frequencies below 11 Hz, triggering is synchronous with the reference, but there are eight triggers per reference signal period. On any given trigger, the X and Y channel outputs and one of the four auxiliary **ADC** inputs is converted. Thus any given auxiliary **ADC** input is converted once every four trigger cycles (i.e. twice per reference cycle). The eight X-channel and eight Y-channel output readings are digitally averaged and it is these average values that are displayed and made available for transfer to a computer.

At frequencies above 11 Hz, the trigger mode automatically changes to the asynchronous trigger mode, but 8F triggering is automatically restored if the frequency subsequently drops to less than 11 Hz.

3.3.17 Choice of Trigger Mode

Trigger modes other than Asynchronous are sometimes used in situations where the random noise level is low and the output time constant is less than, or not much greater than, the reciprocal of the reference frequency. In this situation the following points may be taken into account:

- 1) The Asynchronous trigger mode gives a noisy output which includes beats between the reference frequency and the sampling frequency or its harmonics. This effect is greatly reduced by using the 12 dB/octave slope option rather than 6 dB/octave.
- 2) The Reference trigger mode does not give rise to extra noise, but the measured value of the signal depends on the phase setting in a way that may be unexpected. This results from the fact that there is not more than one sample in each reference cycle, occurring always at the same point in the cycle.
- 3) If the reference frequency is less than 11 Hz, the 8F trigger mode usually gives good results.

3.3.18 Choice of ADC Sample Rate

The inherent problem with any sampling system is the possibility of aliasing with interfering signals. To avoid problems of intermodulation with harmonics of the supply line frequency the ADC sample rate is factory set to 200 Hz for delivery to countries with 60 Hz line frequency, and to 167 Hz for delivery to countries with 50 Hz line frequency. In some situations where high-level interfering signals at other frequencies are causing beats or offsets, the problem can be eliminated by adjustment of the system sample rate either from the front panel (Function 6, Section 3.4.04) or by means of the SAMPLE command.

3.3.19 Output Processor - General

Although shown on the block diagram as a separate entity, the output processor is in fact part of the instrument's main microprocessor. It provides more digital filtering of the X channel and Y channel signals if this is needed in addition to that already performed by the main analog output filters, as well as calculating the vector magnitude and phase of the input signal, the noise, and the ratio of the X-channel output to a voltage applied to the rear-panel **ADC 1** connector.

3.3.20 Output Processor - Vector Magnitude & Phase

The processor implements the magnitude and signal phase calculation. If the input signal $V_s(t)$ is a reference frequency sinusoid of constant amplitude, and the output filters are set to a sufficiently long time constant, the demodulator outputs are constant levels V_x and V_y . The function $\sqrt{(V_x^2 + V_y^2)}$ is dependent only on the amplitude of the required signal $V_s(t)$ (i.e. it is not dependent on the phase of $V_s(t)$ with respect to the reference input) and is computed by the output processor and known as the magnitude output. The phase angle between $V_s(t)$ and the X demodulation function is called the signal phase: this is equal to the angle of the complex quantity $(V_x + jV_y)$ (where j is

the square root of -1) and is also computed by the processor by means of a fast arc tan algorithm.

The magnitude and signal phase outputs are used in cases where phase is to be measured, or alternatively where the magnitude is to be measured under conditions of uncertain or varying phase.

One case of varying phase is that in which the reference input is not derived from the same source as that which generates the signal, and is therefore not exactly at the same frequency. In this case, if the input signal is a sinusoid of constant amplitude, the X channel and Y channel demodulator outputs show slow sinusoidal variations at the difference frequency, and the magnitude output remains steady.

However, the magnitude output has disadvantages where significant noise is present at the outputs of the demodulator. When the required signal outputs (i.e. the mean values of the demodulator outputs) are less than the noise, the outputs take both positive and negative values but the magnitude algorithm gives only positive values: this effect, sometimes called noise rectification, gives rise to a zero error which in the case of a Gaussian process has a mean value equal to 0.798 times the combined root-mean-square (rms) value of the X and Y demodulator noise. Note that unlike other forms of zero error this is not a constant quantity which can be subtracted from all readings, because when the square root of the sum of the squares of the required outputs becomes greater than the total rms noise then the error due to this mechanism disappears.

A second type of signal-dependent error in the mean of the magnitude output occurs as a result of the inherent non-linearity of the magnitude formula: this error is always positive and its value, expressed as a fraction of the signal level, is half the ratio of the mean-square value of the noise to the square of the signal.

These considerations lead to the conclusion that when the magnitude output is being used, the time constants of the demodulator should be set to give the required signal/noise ratio at the X channel and Y channel demodulator outputs; improving the signal/noise ratio by averaging the magnitude output itself is not to be recommended.

For analogous reasons, the magnitude function also shows signal-dependent errors when zero offsets are present in the demodulator. For this reason, it is essential to reduce zero offsets to an insignificant level (usually by the use of the Auto-Offset function) when the magnitude output is to be used.

Note that the majority of signal recovery applications are scalar measurements, where the phase between the required signal and the reference voltage is constant apart from possible phase reversals corresponding to changes in the sign of the quantity being measured. In this situation the lock-in amplifier is used in the normal X-Y mode, with the phase shifter adjusted to maximize the X output and to bring the mean Y output to zero. (Refer to section 3.3.27 for further information on the correct use of the Auto-Phase function for this purpose.)

3.3.21 Output Processor - Noise Measurements

The noise measurement facility uses the output processor to calculate the root mean square value of the output of the X channel.

Because of dynamic range requirements, the quantity actually measured is the mean value of the modulus of X. The result is presented as a percentage of full-scale sensitivity with a scaling factor which gives the correct root-mean-square value if the amplitude distribution is sinusoidal. For the more likely case of a Gaussian distribution, the correct root-mean-square value is obtained by multiplying the reported noise value by a correction factor 1.3313 if the band-pass filter mode is selected, or 1.1986 in the Flat filter mode where a significant contribution to the measured noise is made through windows in the frequency domain centered on harmonics of the reference frequency.

The average is over 1024 samples, taken at a rate equal to the ADC sample rate (Section 3.3.18) and the value of the measured noise output is updated at intervals corresponding to 64 samples.

Note that the settling time of the measured noise output, after a change in the instrument settings or a change in the mean value of X, is $1024/(\text{Sample Rate})$ seconds.

The noise output scaling factor is such that when the X channel output is a steady DC voltage (no noise) the value of the noise output is equal to the value of the X channel output when set to the **% FS/X Y** mode. This is a useful way of checking the system calibration when noise measurements are to be made.

Before measuring the output noise, it is normally necessary to null the mean output of the X channel by implementing a suitable phase shift or by use of the X offset facility.

If the purpose of the measurement is to find the spectral density of a broadband random noise source, and assuming that the low-frequency ranges are not in use, the output time constant should typically be set to 10 ms or 1 ms. The band-pass filter option should normally be used so that the spectral density is measured at a well-defined frequency, i.e. the reference frequency; the use of this filter is also important in reducing the effect of interfering signals, but at lower reference frequencies it may have a significant effect on the overall noise bandwidth. When setting the reference frequency, the power line frequency and its harmonics should be avoided, and a useful precaution is to experiment with small changes in the reference frequency to ensure that the measurement is not being affected by an external interfering signal causing a spurious peak to occur in the measured noise spectrum.

The equivalent noise bandwidth of the output filter is either 1/4TC (6 dB/octave slope) or 1/8TC (12 dB/octave slope) where TC is the selected time-constant.

Because of possible inaccuracies in calculated values of noise bandwidth, the user is strongly recommended to calibrate spectral density measurements by the use of a resistor connected to the input as a noise voltage source. For measurements at frequencies up to a few kHz a value of 100 k Ω is commonly used, giving 40.7 nV/ $\sqrt{\text{Hz}}$

at 300 K. At higher frequencies the input capacitance of the lock-in makes it desirable to use a lower resistor value, but in this case the voltage noise of the lock-in must be taken into account. Note that this method of calibration is appropriate only to those instruments which, like the 5110, have a JFET input stage.

The user is strongly advised to use an oscilloscope attached to the front-panel **SIG MON** (signal monitor) output when making noise measurements as this is the best way of ensuring that one is measuring a random process rather than line pick-up or other non-random signals.

3.3.22 Analog Panel Meter

The front-panel analog meter is always driven by the output of an internal digital to analog converter, and can display the X-channel, the signal magnitude, the noise or the ratio outputs. The meter current is always derived from a digital to analog converter in order to take advantage of the different ADC sampling modes provided.

3.3.23 CH1 and CH2 Analog Outputs

The two front-panel connectors **CH1** and **CH2** give analog voltages proportional to the instrument outputs, depending on the display mode set in the **DISPLAY 2** section.

When set to the **% FS/X Y** or **X Y** display modes, **CH1** and **CH2** give analog voltages proportional to the X-channel and Y-channel outputs respectively. With time-constant values of 30 ms or shorter, they are obtained directly from the output filters, but at higher values, they are generated from the output processor via a digital-to-analog converter. The full-scale voltage is ± 1.000 V although the outputs remain valid up to ± 1.200 V

When set to the **% FS/R θ** or **R θ** mode or one of the **SPEC** (special) modes, the voltages at the **CH1** and **CH2** connectors depend on the mode selected, and are generally derived from the output processor via a digital-to-analog converter.

3.3.24 Auxiliary Analog Inputs and Output (ADCs and DAC)

The model 5110 incorporates four auxiliary ADC inputs, which are digitized by a conventional sampled converter giving a resolution of 1 mV in ± 12.000 V. These inputs can be used for digitizing slowly changing or DC signals which are associated with an experiment, such as those generated by temperature and pressure transducers, so that they can be incorporated in ratio calculations or transferred to a controlling computer. The sample rate depends on the main ADC trigger mode, as discussed already in section 3.3.16

Two auxiliary DAC analog outputs are also provided which offer the same resolution as the ADCs, namely 1 mV in ± 12.000 V.

3.3.25 Overload Detection

There are three overload detectors in the signal channel, at the following locations in the circuit:

1. before the line frequency rejection filter;
2. before the main signal-channel filter;
3. at the output of the main signal-channel filter, i.e. the same point as accessed by the **SIG MON** connector.

The outputs of these detectors are connected to the **OVLD** indicator in the **SENSITIVITY** section, and when this lights the unit is in input overload. Such overloads are removed by increasing the full-scale sensitivity value, by increasing the value of the dynamic reserve mode selection, or by introducing one or more of the signal-channel filters.

There are two overload detectors following the demodulators, located at the inputs to the X-channel and Y-channel analog-to-digital converters. The outputs of these detectors are connected to the **OVLD** indicator in the **DISPLAY 2** section, and when this lights the unit is in output overload. If overload occurs at either of these points with less than 120 percent full-scale showing at either the X or the Y outputs, the overload can be removed by increasing the time constant of the output filters; by increasing the value of the dynamic reserve or by making use of the signal-channel filters.

Input and output overload conditions cause the assertion of bit 4 in the status byte, and one or more of bits 3, 4, 5 and 6 in the overload status byte. Hence a controlling computer program can determine the location of the overload and make the required adjustments to remove it.

3.3.26 Main Microprocessor - General

All functions of the instrument are under the control of a microprocessor which in addition drives the front panel displays, processes front panel key operations and supports the RS232 and GPIB (IEEE-488) computer interfaces. The processor also drives the instrument's rear-panel 8-bit digital (TTL) programmable output port which may be used for controlling auxiliary apparatus.

3.3.27 Main Microprocessor - Auto Functions

The microprocessor also controls the instrument's auto-functions, which are control operations executed by means of a single command or two key-presses. These functions allow easier, faster operation in most applications, although direct manual operation or special purpose control programs may give better results in certain circumstances. During application of several of the auto-functions, decisions are made on the basis of output readings made at a particular moment. Where this is the case, it is important for the output time constant set by the user to be long enough to reduce the output noise to a sufficiently low level so that valid decisions can be made and that sufficient time is allowed for the output to settle.

The following sections contain brief descriptions of the auto-functions. Note that unlike the auto-functions built into some instruments the functions in the 5110 operate once only each time they are selected, rather than remaining active. Hence, for example, the auto-sensitivity function does not cause the instrument to continuously adjust its full-scale sensitivity as the input signal level changes, but rather makes one

adjustment which will attempt to match the signal level at the time the function is called.

Auto-Sensitivity

The auto-sensitivity function causes the sensitivity to be automatically set to the sensitivity range that yields a signal magnitude output between 20% and 90% of full scale. Note that the auto-sensitivity function does not depend on the phase setting to work correctly. However, the phase will have to be set correctly to maximize the X-output reading.

In the presence of noise, or a time-varying input signal, it may be a long time before the Auto-Sensitivity sequence comes to an end, and the resulting setting may not be what is really required.

Auto-Phase

In an Auto-Phase operation the value of the signal phase is computed and an appropriate phase shift is then introduced into the reference channel so as to bring the value of the signal phase to zero. The intended result is to null the output of the Y-channel while maximizing the output of the X-channel.

Any small residual phase can normally be removed by calling Auto-Phase for a second time after a suitable delay to allow the outputs to settle.

The Auto-Phase facility is normally used with a clean signal which is known to be of stable phase. It usually gives very good results provided that the X-channel and Y-channel outputs are steady when the procedure is called.

If a zero error is present on the outputs, such as may be caused by unwanted coupling between the reference and signal channel inputs, then the following procedure should be adopted:-

- 1) Remove the source of input signal, without disturbing any of the connections to the signal input which might be picking up interfering signals from the reference channel. In an optical experiment, for example, this could be done by shielding the detector from the source of chopped light.
- 2) Execute an Auto-Offset operation, which will reduce the X-channel and Y-channel outputs to zero.
- 3) Re-establish the source of input signal. The X-channel and Y-channel outputs will now indicate the true level of input signal, *at the present reference phase setting*.
- 4) Execute an Auto-Phase operation. This will set the reference phase shifter to the phase angle of the input signal. However, because the offset levels which were applied in step 2 were calculated at the original reference phase setting, they will not now be correct and the instrument will in general display a non-zero Y-channel output value.
- 5) Remove the source of input signal again.

- 6) Execute a second Auto-Offset operation, which will reduce the X-channel and Y-channel outputs to zero at the new reference phase setting.
- 7) Re-establish the source of input signal.

This technique, although apparently complex, is the only way of removing the effect of crosstalk which is not generally in the same phase as the required signal.

Auto-Offset

In an Auto-Offset operation the X offset and Y offset functions are turned on and are automatically set to the values required to give zero values at both the X and the Y outputs. Any small residual values can normally be removed by calling Auto-Offset for a second time after a suitable delay to allow the outputs to settle.

The primary use of the Auto-Offset is to cancel out zero errors which are usually caused by unwanted coupling or crosstalk between the signal channel and the reference channel, either in the external connections or possibly under some conditions in the instrument itself. Note that if a zero error is present, the Auto-Offset function should be executed before any execution of Auto-Phase.

Auto-Measure

The auto-measure function causes the following actions to occur:-

- 1) The signal channel filter is set to bandpass mode, and tuned to the reference frequency.
- 2) The display is set to **% FS/X Y**
- 3) Output offsets and output expand are turned off.
- 4) Normal dynamic reserve is selected.
- 5) The time constant is set to 100 ms and the slope to 12 dB/octave.
- 6) An auto-sensitivity function is performed.
- 7) An auto-phase operation is performed.

The Auto-Measure function is intended to give a quick setting of the instrument which will be approximately correct in typical simple measurement situations. For optimum results in any given situation, it may be convenient to start with Auto-Measure and to make subsequent modifications to the settings of individual controls.

Auto Tune

If the manual filter tuning mode is active, pressing the auto-tune operation sets the filter frequency to the reference frequency (in 2F mode to twice the reference frequency). If the tracking filter tuning mode is in effect then auto-tune has no effect.

This completes the description of the main functional blocks of the instrument.

3.4 General

3.4.01 Accuracy

When the demodulator is operating under correct conditions, the absolute gain accuracy of the instrument is limited by the analog components in the signal channel and output channels, and the absolute phase accuracy is limited by the analog components in both the signal and output channels and in the reference channel. The resulting typical accuracy in flat mode is ± 1 percent of the full-scale sensitivity and ± 1 degree.

3.4.02 Power-up Defaults

All instrument settings are retained when the unit is switched off. When the instrument is switched on again the settings are restored but with the following exceptions:-

- a) The GPIB mask byte is set to zero.
- b) The REMOTE parameter is set to zero (front panel controls are enabled).
- c) The front-panel lights are turned on.
- d) XYDIRECT mode is turned off.

4.1 Front Panel

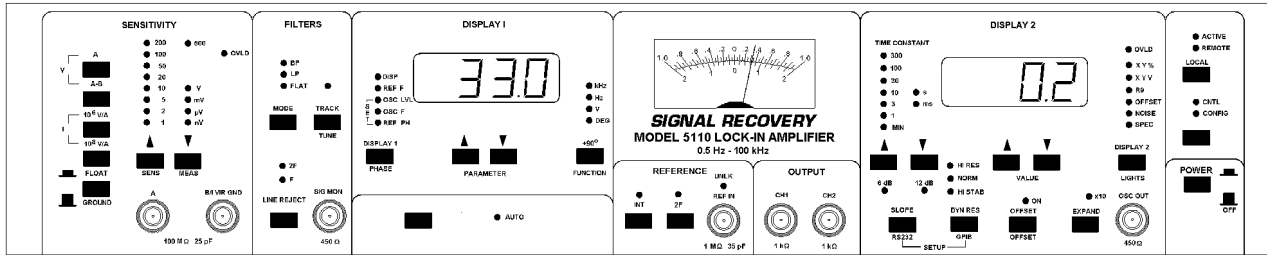


Figure 4-1, Model 5110 Front Panel Layout

The front panel of the Model 5110 follows a logical layout, with controls and displays associated with the input on the left-hand side and those with the output on the right-hand side. As shown in figure 4-1 above, the controls, displays and connectors are organized for convenience into six labeled groups, with additional areas for the analog panel meter, power switch and two other controls.

4.1.01 Connectors

There are seven BNC connectors mounted on the front panel, used for the most common connections to and from the user's experiment. Their purpose is described in the following paragraphs. Connectors used less frequently are mounted on the rear-panel and are described in section 4.2

A and B/I VIR GND BNC connectors

The **A** connector is the signal input connector for use in single-ended and differential voltage mode. The **B/I VIR GND** connector is the signal input connector for use in differential voltage mode (A-B). It is also the signal input connector when current input mode is selected.

SIG MON BNC connector

The signal at this connector is that at the output of the signal channel, immediately prior to the phase sensitive detectors. Because of the operation of the algorithm that distributes the overall gain between the signal channel and the output filters, the gain between the **A** or **A-B** input connectors and this connector depends on a number of factors. These include the full-scale sensitivity range, the main signal channel filter setting, the setting of the dynamic reserve control and the setting of the output filter time-constants. This should be taken into account when the using the **SIG MON** output to monitor correct lock-in amplifier operation or to implement a stand-alone tuned amplifier.

The signal monitor output is unbuffered, and has an output impedance of 450 Ω. As such, it should not be connected to loads of less than 15 kΩ. Internal noise and switching spikes may be present. The higher the reserve setting, the greater their amplitude will be relative to that of the signal.

REF IN connector

This is the input connector for a general purpose external reference signal. The instrument is fitted with two reference input connectors for external reference mode operation, **TTL REF IN** (rear panel) and **REF IN** (front panel). Wherever possible, use the rear-panel TTL input, particularly if the reference signal is asymmetrical. If the available reference is not suitable for driving a TTL input, then use the front-panel reference input.

The input impedance at the front-panel **REF IN** connector is 1 M Ω in parallel with 30 pF.

CH1 and CH2 connectors

The signals at these two connectors are analog voltages representing the instrument outputs, such as X and Y channel outputs, or vector magnitude and phase. The actual output available at each connector depends on the setting of the **DISPLAY 2** section **DISPLAY/LIGHTS** control.

The output impedance at each of these connectors is 1 k Ω .

OSC OUT connector

This is the output connector for the internal oscillator, which may be set to generate a sine wave between 0.5 Hz and 120 kHz with amplitude between 1 mV and 1.999 V or 5.000 V rms.

The output impedance of the oscillator is nominally 450 Ω , giving a useful 10:1 attenuation when loaded with 50 Ω . Note that when using the output with a typical cable and a high-impedance load, phase shifts of a few degrees may occur at higher frequencies.

4.1.02 Keys

The instrument is fitted with twenty-nine keys of four different types, as follows:-

Voltage/Current mode Input Keys

The upper four keys at the extreme left-hand side of the front-panel are used to select the input mode and are mechanically interlocked, such that pressing one of them results in the release of another. At least one of these keys should always be depressed.

On/Off keys

Some keys, such as the **POWER** switch and the **EXPAND** key have an alternate On-Off operation each time they are pressed. The present state of the key is indicated either by the fact that the position of the key relative to the panel differs for each state, or by an adjacent indicator being lit or unlit.

Selection keys

Controls with several possible settings, such as the signal channel filter **MODE** control, change setting each time the corresponding key is pressed. Hence, for example, to change the filter from flat to bandpass mode, press the filter **MODE** key twice; to change it back to flat mode, press it once more.

▲ and ▼ keys

There are four pairs of ▲ and ▼ keys which are used to adjust those controls, such as input sensitivity or internal oscillator frequency, which need to be changed over a wide range of values.

The ▲ and ▼ keys corresponding to the input sensitivity and output time constant controls are used to set these to the required values simply by repeatedly pressing the relevant key until the required setting is illuminated; press the ▲ key once to decrease the sensitivity, or increase the time constant, by one step, and press the ▼ key once to increase the sensitivity, or decrease the time constant, by one step.

The ▲ and ▼ keys immediately under the **DISPLAY 1** LED display adjust the internal oscillator frequency and amplitude, the signal channel filter frequency and the reference phase. Each control is changed by first using the **DISPLAY 1** select key to choose the required control, at which point the LED will show the present setting of the control. Then, use the ▲ and ▼ keys to change the setting. Note that the keys are fitted with an acceleration function, such that if they are pressed and released quickly, the setting will change by one in the least significant digit, but if pressed and held then the setting will change approximately ten times faster. This makes it very easy and fast to change the setting of a control over a wide range of values.

Finally, the ▲ and ▼ keys immediately under the **DISPLAY 2** LED display adjust the output offset level. First use the **DISPLAY 2** section **DISPLAY/LIGHTS** key to select **OFFSET**, at which point the LEDs display the present offset setting. Then use the ▲ and ▼ keys to change the value as required.

Dual Key Functions

Some operations, such as the auto-functions, require that two keys be pressed in succession. For example, in order to perform an auto-measure operation, first press the red **AUTO** key. Then press the ▼ key in the **SENSITIVITY** section which has the secondary **MEAS** (auto-measure) function.

4.1.03 Control Setting Indicators

The present setting of the various instrument controls is shown by an LED adjacent to the relevant legend being lit. The setting of controls, such as input sensitivity, which require both a value and a unit, are indicated by illuminating two LEDs, one for the value and one for the unit.

4.1.04 Overload Indicators

There are two front-panel red overload indicators which show **OVL** under overload conditions. The first, in the top right-hand corner of the **SENSITIVITY** section, indicates an input overload, and the second, in the top right hand corner of the **DISPLAY 2** section, indicates an output overload.

4.1.05 LED Displays

The instrument is fitted with two, 3½ digit, LED displays. The left-hand one, **DISPLAY 1**, is used to adjust three controls (internal oscillator amplitude, internal

oscillator frequency, and reference phase), to display the measured reference frequency, and, in conjunction with **DISPLAY 2**, to display the instrument's outputs.

The right-hand LED, **DISPLAY 2**, is used to adjust the output offset control and to display the instrument's outputs.

4.1.06 Analog Panel Meter

The analog panel meter in the center of the instrument always displays the same output that appears on the **DISPLAY 1** LED when the **DISPLAY 1** mode is set to **DISP**.

4.2 Rear Panel

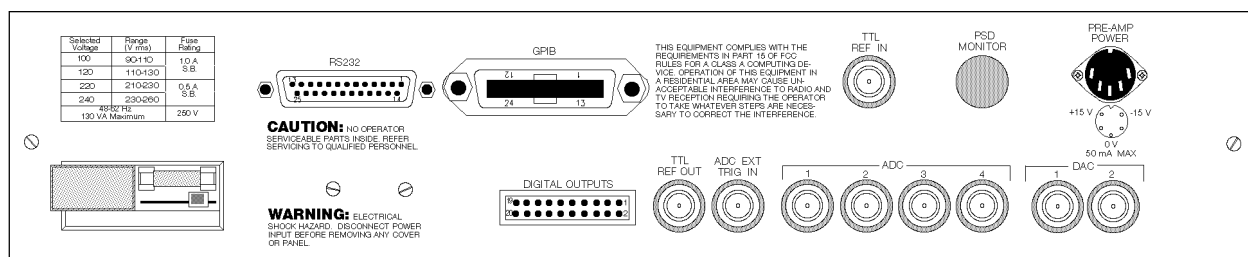


Figure 4-2, Model 5110 Rear Panel Layout

As shown in figure 4-2, the line power input assembly, RS232 and GPIB (IEEE-488) connectors, preamplifier power connector, digital output port connector and nine BNC signal connectors are mounted on the rear panel of the instrument. Brief descriptions of these are given in the following sections.

4.2.01 Line Power Input Assembly

This houses the line voltage selector and line input fuse. To check, and if necessary change, the fuse or line voltage see the procedure in section 2.1.05.

4.2.02 RS232 Connector

This 25-pin D type RS232 interface connector implements pins 1, 2, 3, 4, 7 and 20 (Earth Ground, Transmit Data, Receive Data, Request to Send, Logic Ground and Data Terminal Ready) of a standard DTE interface. To make a connection to a PC-compatible computer, it is normally sufficient to use a three-wire cable connecting Transmit Data to Receive Data, Receive Data to Transmit Data, and Logic Ground to Logic Ground. Appendix D shows the connection diagrams of cables suitable for computers with 9-pin and 25-pin serial connectors. Pinouts for this connector are given in appendix B.

4.2.03 GPIB Connector

The GPIB interface connector conforms to the IEEE-488.1 1978 Instrument Bus Standard. The standard defines all voltage and current levels, connector specifications, timing and handshake requirements.

4.2.04 PREAMP POWER Connector

This connector supplies ± 15 V DC at up to 50 mA and can be used for powering any of several optional remote preamplifiers available from **SIGNAL RECOVERY**. Pinouts for this connector are given in appendix B.

4.2.05 TTL REF IN Connector

This connector is provided to allow TTL compatible pulses to be used as the reference input when operating in external reference mode. Wherever possible, it should be used in preference to the front-panel **REF IN** connector.

4.2.06 DIGITAL OUTPUTS Connector

This connector provides eight TTL output lines, each of which can be set high or low. It is most commonly used for controlling auxiliary apparatus, such as lamps, shutters and heaters. Pinouts for this connector are given in appendix B.

4.2.07 TTL REF OUT Connector

The signal at this connector is a TTL-compatible waveform synchronous with the reference. This output monitors correct reference channel operation but its polarity is not uniquely defined so that it does not necessarily show the correct phase relationship with the front-panel **SIG MON** output.

4.2.08 ADC EXT TRIG IN Connector

This connector accepts a TTL-compatible input and can be used for triggering the auxiliary Analog to Digital Converters (ADCs). The input operates on the positive edge only.

4.2.09 ADC 1 - ADC 4 Connectors

The input voltages at these connectors may be digitized using the auxiliary ADCs and read either from the front panel or by the use of a computer command. The input voltages are sampled and held when the ADC is triggered, and several different trigger modes are available. These modes can be set either from the front panel or by using a remote computer command. The input voltage range is ± 12.0 V and the resolution is 1 mV.

The voltage at the **ADC 1** input is used as the denominator for the ratio calculation when this output is selected.

4.2.10 DAC 1 and DAC 2 Connectors

These are the output connectors for the instrument's auxiliary digital to analog converter. The voltage at these connectors can be set either from the front panel or by the use of a remote computer command in the range ± 12.0 V to a resolution of 1 mV. The outputs are of low impedance.

5.1 Introduction

This chapter describes how to operate the model 5110 using the front panel controls, and discusses its capabilities when used in this way. Chapter 6 provides similar information in the situation where the unit is operated remotely using one of the computer interfaces.

The model 5110 uses a combination of keys and LED indicators on its front panel to set the various instrument controls, allowing the operator to determine the instrument's status at a glance. Controls, associated indicators, displays and connectors are grouped for operator convenience into clearly delineated areas. There are six such labeled areas: **SENSITIVITY**, **FILTERS**, **DISPLAY 1**, **REFERENCE**, **OUTPUT**, and **DISPLAY 2**, as well as the analog panel meter, power switch, and, in the upper right-hand corner, the **LOCAL** and **CONFIG/CNTL** keys.

Specific information regarding each functional grouping is provided in the following sections.

5.2 SENSITIVITY Group

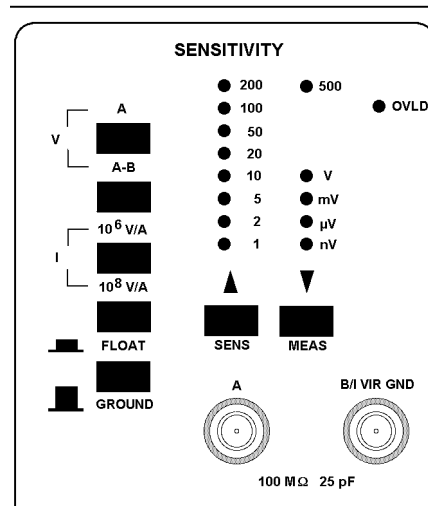


Figure 5-1, SENSITIVITY Group

As shown in figure 5-1, the **SENSITIVITY** group contains the two input connectors, several keys and an overload indicator. Information about each follows.

5.2.01 A and B/I VIR GND BNC Connectors

The function of these input connectors depends on whether the 5110's input is operating in the voltage-sensitive or current-sensitive mode, as set by the upper four right-hand interlocking keys. Pressing either **A** or **A-B** results in a voltage-sensitive input. Pressing **10⁶ V/A** or **10⁸ V/A** gives a current sensitive input.

5.2.02 A Key

When this key is depressed, the input is in single ended voltage mode, and the signal to be measured should be connected to the BNC connector marked **A**.

5.2.03 A-B Key

When this key is depressed, the input is in differential voltage mode, and the differential voltage to be measured should be connected to the BNC connectors marked **A** and **B/I VIR GND**.

5.2.04 10^6 V/A and 10^8 V/A Keys

When either of these keys are depressed, the input is in current sensitive mode and the signal to be measured should be connected to the virtual-ground BNC connector marked **B/I VIR GND**. When 10^6 V/A is selected, an applied current of 10^{-6} A will give 1 V at the input of the voltage preamplifier. When 10^8 V/A is selected, an applied current of 10^{-8} A gives 1 V at the input of the voltage preamplifier. The preamplifier's sensitivity in turn is set by the full-scale sensitivity control. Thus the overall sensitivity in current measurements is influenced by both the selected current conversion gain (10^6 V/A or 10^8 V/A) and the selected full-scale voltage sensitivity. The phasing at the **B/I VIR GND** connector in current measurement operation is the same as that at the **A** connector in voltage measurement operation.

5.2.05 FLOAT/GROUND Key

When this key is depressed, the shells of both input connectors are returned to chassis ground through a 1 k Ω resistor for improved ground-loop interference rejection. When released, the shells of the input connectors are returned directly to chassis ground.

5.2.06 SENSITIVITY $\blacktriangle$ and $\blacktriangledown$ Keys

The principal function of the **SENSITIVITY** $\blacktriangle$ and $\blacktriangledown$ keys is to set the instrument's full-scale sensitivity to values in the range 100 nV to 1 V rms in a 1-2-5-10 sequence. The present setting is shown by the relevant figure and units indicators being lit.

Both keys also have secondary auto-functions, as indicated by the label beneath each of these keys. The secondary function is accessed when the red **AUTO** key is pressed prior to pressing the arrow key.

Auto-Sensitivity

The secondary function of the **SENSITIVITY** $\blacktriangle$ key is to initiate an auto-sensitivity operation. If the red **AUTO** key (located beneath the **DISPLAY 1** group) has been pressed, pressing the **SENSITIVITY** $\blacktriangle$ key will cause the sensitivity to be automatically set to the sensitivity range that yields a signal magnitude output between 20% and 90% of full scale. Note that the auto-sensitivity function does not depend on the phase setting to work correctly. However, the phase will have to be set correctly to maximize the X-output reading.

Auto-Measure

The secondary function of the **SENSITIVITY** ▼ key is to initiate an auto-measure operation, which causes the following actions to occur:-

- 1) The signal channel filter is set to bandpass mode, and tuned to the reference frequency.
- 2) The display is set to **% FS/X Y**
- 3) Output offsets and output expand are turned off.
- 4) Normal dynamic reserve is selected.
- 5) The time constant is set to 100 ms and the slope to 12 dB/octave.
- 6) An auto-sensitivity function is performed.
- 7) An auto-phase operation is performed.

The Auto-Measure function is intended to give a quick setting of the instrument which will be approximately correct in typical simple measurement situations. For optimum results in any given situation, it may be convenient to start with Auto-Measure and to make subsequent modifications to the settings of individual controls.

5.2.07 OVLD Indicator

The **SENSITIVITY** group red **OVLD** indicator lights when overload occurs anywhere in the signal channel, including both pre-filter and post-filter locations. Should this happen, the overload condition must be corrected before valid measurements can be taken. Appropriate actions include changing the dynamic reserve setting, reducing the signal-channel full-scale sensitivity setting, activating the signal channel filter and inserting the line frequency rejection filter.

5.3 FILTERS Group

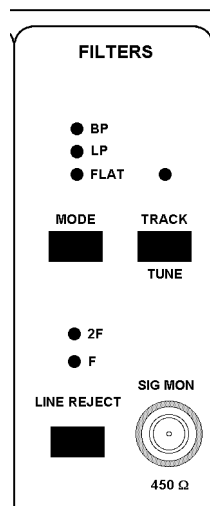


Figure 5-2, FILTERS Group

The keys in this area, shown in figure 5-2, determine the behavior of the main signal channel filter and the line frequency rejection filter. In addition, the output of the signal channel following the input amplifier and filtering stages and prior to the phase sensitive detectors is provided at the **SIG MON** connector.

5.3.01 MODE Key

This key allows the main signal channel filter to be set to one of four characteristics, flat, notch, low-pass and bandpass. Each time the key is pressed the mode changes, so that for example to change from flat to notch mode, press the key once; to change back to flat mode, press the key twice more.

A brief description of each filter mode follows.

FLAT Mode

No filtering occurs with this selection and the PSDs (phase sensitive detectors) are switched to square-wave drive, giving the instrument an overall “square wave” response. In other words, the unit will detect input signals not only at the reference frequency but also at its odd harmonics, with the response to the “ n ”th harmonic being $1/n$ times that at the fundamental.

As a result, for the same amplitude input square wave, the output will be about 23% higher with **FLAT** selected than with **BP** (bandpass) or **LP** (low-pass) selected. With the latter choices, only the fundamental frequency component of the square wave contributes to the output.

BP (bandpass) and LP (low-pass) Modes

When the bandpass or low-pass modes are selected, the PSDs are switched to Walsh-function drive. Consequently the overall response of the instrument becomes fundamental (sinusoidal) since signals at the third and fifth harmonics of

the reference frequency are rejected by the PSDs, whilst higher harmonics are rejected by the signal channel filter. These settings give the maximum dynamic reserve.

The low-pass filter gives maximum rejection of the higher harmonics but allows frequencies below the set frequency to be passed unattenuated to the PSD. The bandpass filter attenuates both above and below the set frequency but with less harmonic rejection than would be provided by the low-pass filter.

5.3.02 TRACK/TUNE Key

This key is primarily used to switch the main filter frequency tracking function on and off. Its secondary function is to initiate an auto-tune operation.

To set the filter frequency to a particular value, set the reference frequency to the required value and then press **TRACK/TUNE**, so the indicator above it is lit. Then press **TRACK/TUNE** again. The filter frequency will remain at the established setting, in spite of any subsequent changes to the reference frequency.

Continuous tracking is implemented by pressing the **TRACK/TUNE** key once, so that the indicator above it is lit. To avoid noise resulting from tracking small random frequency changes, do not use continuous tracking in situations where the reference frequency is nominally constant.

If the manual tuning mode is active, pressing the **AUTO** key and then the **TRACK/TUNE** key causes an auto-tune operation to occur, which sets the filter frequency to the reference frequency (in 2F mode to twice the reference frequency). If the tracking mode is selected then auto-tune has no effect.

5.3.03 LINE REJECT Key

This key controls the line frequency rejection filters. The choices are:

- 1) No line filter selected (neither **F** nor **2F** lit).
- 2) **F** (notch tuned to the line frequency; **F** lit).
- 3) **2F** (notch tuned to twice the line frequency; **2F** lit.)
- 4) **F** and **2F** (two notch filters active, one tuned to the line frequency and the other to twice the line frequency; both **F** and **2F** lit.)

The line filters are completely independent of the tuned filter functions previously described. The **F** filter attenuation at the line frequency is more than 34 dB at a stopband of $\pm 1\%$ of the nominal filter frequency.

The line frequency filter is supplied set to match the line frequency of the country of destination, but it will be appreciated that if a unit is moved from a 50 Hz area to a 60 Hz area then the filter will need to be adjusted. This can be done via one of the computer interfaces using the **LINE** command.

5.3.04 SIG MON BNC Connector

The signal at this connector is that at the output of the signal channel, immediately prior to the phase sensitive detectors. Because of the operation of the algorithm that distributes the overall gain between the signal channel and the output filters, the gain between the **A** or **A-B** input connectors and this connector depends on a number of factors. These include the full-scale sensitivity range, the main signal channel filter setting, the setting of the dynamic reserve control and the setting of the output filter time-constants. This should be taken into account when using the **SIG MON** output to monitor correct lock-in amplifier operation or to implement a stand-alone tuned amplifier.

The signal monitor output is unbuffered, and has an output impedance of 450 Ω . As such, it should not be connected to loads of less than 15 k Ω . Internal noise and switching spikes may be present. The higher the reserve setting, the greater their amplitude will be relative to that of the signal.

5.4 DISPLAY 1 Group

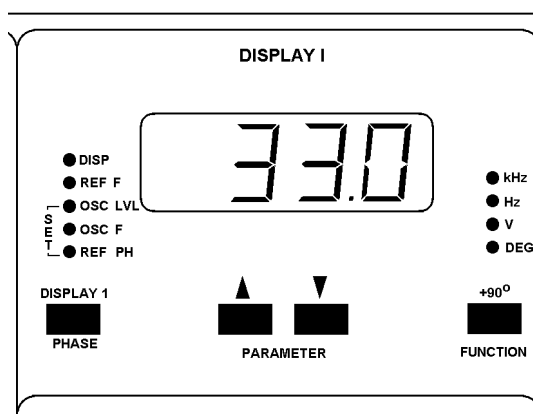


Figure 5-3, DISPLAY 1 Group

The **DISPLAY 1** group, shown above in figure 5-3, contains a 3½ digit LED display used to set three instrument controls, to display the present reference frequency and to function as one of two output displays. In this last mode, the value displayed on the meter is determined by the **DISPLAY 2** selection, described later in the **DISPLAY 2** discussion (see section 5.10).

5.4.01 DISPLAY 1 /PHASE Key

This key is primarily used to specify what is displayed on the **DISPLAY 1** group LED display but it also has a secondary purpose, which is to initiate an auto-phase operation.

Each time the **DISPLAY 1/PHASE** key is pressed, the value shown on the **DISPLAY 1** LED changes, with the present setting being shown by the relevant indicator being lit, as follows:

SET REF PH (Set reference phase)

When the **SET REF PH** indicator is lit, the LED displays the present setting, in degrees, of the reference channel phase shifter. The setting can be changed using the ▲ and ▼ **PARAMETER** keys immediately below the LED. The step size is 5, 25 or 100 millidegrees as set by function 2 (see discussion of **FUNCTION** key operation in section 5.4.03)

SET OSC F (Set internal oscillator frequency)

When the **SET OSC F** indicator is lit, the LED displays the present setting, in Hz or kHz, of the internal oscillator frequency. The setting can be changed using the ▲ and ▼ **PARAMETER** keys immediately below the LED.

***NOTE:** Because the internal oscillator error may be $\pm 1\%$, the actual oscillator frequency obtained may not necessarily agree with the value shown in the OSF F display.*

SET OSC LVL (Set internal oscillator level)

When the **SET OSC LVL** indicator is lit, the LED displays the present setting, in volts rms, of the internal oscillator output amplitude. The setting can be changed between zero and 1.999 V rms using the ▲ and ▼ **PARAMETER** keys immediately below the LED. Note also that it is possible to set the amplitude to 5.000 V, but only by using the OA computer command. In the external reference mode the oscillator is independent of the reference channel and can be used for any purpose.

REF F

When the **REF F** indicator is lit, the LED displays the measured reference frequency, in Hz or kHz. Reference lock need not be established. As long as a valid reference input signal is present (either internal or external), a valid frequency reading is obtained.

DISP

The **DISP** position allows measurement and setting functions that depend on one parameter being displayed in **DISPLAY 1** and another in **DISPLAY 2**. Included are all of the output display functions and offset channel selection. See the discussions of the various **DISPLAY 2** functions in section 5.9.08 for information about the significance of the **DISPLAY 1** indication when **DISP** is selected.

Normally, unless a **DISPLAY 1** control that requires a selection other than **DISP** is being adjusted, set the **DISPLAY 1** selection to **DISP**.

The secondary function of the **DISPLAY 1/PHASE** key is to initiate an auto-phase operation. If the red **AUTO** key (located beneath the **DISPLAY 1** group) has been pressed, pressing the **DISPLAY 1/PHASE** key will cause the reference phase to be adjusted for maximum X-channel output.

5.4.02 Display 1 ▲ and ▼ PARAMETER Keys

As already mentioned above, the ▲ and ▼ **PARAMETER** keys are used to adjust the selected control.

5.4.03 +90°/FUNCTION Key

The primary purpose of this key is to add 90° to the present setting of the reference phase shifter each time it is pressed, for fast four-quadrant coverage. With the reference signal connected and proper reference channel operation established, the phase is normally adjusted to bring the signal and reference into phase at the X-channel PSD. This will result in maximum positive indication on the **DISPLAY 1** LED and minimum indication on the **DISPLAY 2** LED (**DISPLAY 1** selection **DISP** and **DISPLAY 2** selection **% FS/X Y** or **X Y**). Maximum negative indication on the **DISPLAY 1** LED will occur if the signal and phase are 180° out of phase. (**DISPLAY 2** LED still a minimum)

The secondary purpose of the key is in conjunction with the **CONFIG/CNTL** key, described later in section 5.10.04. If the **CONFIG/CNTL** key followed by the **+90°/FUNCTION** key are pressed then the instrument enters the function mode which allows the setting of number of special controls.

When the function mode is active, which is indicated by the **CNTL** indicator being lit, a number identifying the function, or control, appears in the **DISPLAY 1** LED. This number, and hence the selected function, can be adjusted using the **PARAMETER** ▲ and ▼ keys located below the **DISPLAY 1** LED, while the setting of the selected function is adjusted using the **VALUE** ▲ and ▼ keys located below the **DISPLAY 2** LED. The six controls accessed in this way are:

Function 1: ADC Conversion (Trigger) Mode Control

When the function mode is active and the digit “1” appears on the **DISPLAY 1** LED, the analog-to-digital converter (ADC) trigger mode may be adjusted. This sets the trigger mode for the instrument’s ADC converter, which digitizes the X and Y channel outputs and the voltages applied to the rear-panel **ADC 1 - ADC 4** connectors.

There are five different trigger modes, indicated by the code numbers 0, 1, 2, 4 and 8, with the present setting shown on the **DISPLAY 2** LED. The setting can be changed using the **VALUE** ▲ and ▼ keys located below this display. The five settings are as follows:-

Function 1: Setting “0”: Reference Synchronous Trigger

In this setting, the ADC is triggered synchronously with the reference frequency. The maximum conversion rate is nominally once every 5 ms (200 Hz), so that at low reference frequencies there will be one trigger per reference period. At higher reference frequencies there will be one trigger per some multiple of the reference period.

Each trigger results in the conversion of X- and Y-channel outputs and one of the four auxiliary **ADC** inputs. Thus any given **ADC** is converted once every four trigger cycles.

Function 1: Setting “1”: External Trigger Mode

In this setting the ADC is triggered once on each positive going TTL transition detected at rear-panel **ADC EXT TRIG IN** connector. If an applied trigger occurs before the converter has finished processing the previous trigger, it is ignored. On any given trigger, the X- and Y-channel outputs and all four of the auxiliary **ADC** inputs are converted.

Function 1: Setting “2”: Asynchronous Trigger Mode

In this setting the ADC is triggered asynchronously at the preset system sample rate, which is adjustable over the range 100 Hz to 200 Hz by means of a front panel control or remote computer command. On any given trigger, the X- and Y-channel outputs and one of the four auxiliary **ADC** inputs are converted. Thus any given auxiliary **ADC** input is converted once every four trigger cycles (i.e. at a quarter of the system sample rate).

Function 1: Setting “4”: Ratio Trigger Mode

In this setting, triggering is synchronous with the reference signal as for mode 0, but only the X- and Y- channel outputs and the **ADC 1** input are converted. The maximum conversion rate is nominally once every 5 ms (200 Hz), so that at low reference frequencies there will be one trigger per reference period. At higher reference frequencies there will be one trigger per some multiple of the reference period.

The advantage of this mode over the reference synchronous trigger mode (mode 0) is that the **ADC 1** input is converted at the same rate as the X- and Y-channel outputs, giving better results when the instrument is set to the ratio display mode.

NOTE: *In the ratio trigger mode, the ADC 2, 3 and 4 inputs are never converted.*

Function 1: Setting “8”: “8F” Trigger Mode

In this setting, at reference frequencies below 11 Hz, triggering is synchronous with the reference, but there are eight triggers per reference signal period. On any given trigger, the X and Y channel outputs and one of the four auxiliary **ADC** inputs is converted. Thus any given auxiliary **ADC** input is converted once every four trigger cycles (i.e. twice per reference cycle). The eight X-channel and eight Y-channel output readings are digitally averaged and it is these average values that are displayed and made available for transfer to a computer.

At frequencies above 11 Hz, the trigger mode automatically changes to the asynchronous trigger mode, but 8F triggering is automatically restored if the frequency is subsequently drops to less than 11 Hz.

Function 2: Reference Phase Shifter Resolution

When the function mode is active and the digit “2” appears on the **DISPLAY 1** LED, the phase-setting resolution of the reference phase shifter to be set to 100, 25 or 5 millidegrees. The present setting is shown on the **DISPLAY 2** LED and can be changed using the **VALUE** ▲ and ▼ keys located below this display. The three settings are as follows:

Function 2: Setting “100”: 100 m° Phase Setting Resolution

In this setting, the phase setting resolution is 100 m°, which is compatible with the front-panel reference phase shifter display resolution and is suitable for most measurements.

Function 2: Setting “25”: 25 m° Phase Setting Resolution

In this setting, the phase setting resolution is 25 m°

Function 2: Setting “5”: 5 m° Phase Setting Resolution

In this setting, the phase setting resolution is 5 m°, which is suitable for use in applications where it is necessary to obtain a "perfect" null of the Y-channel output. Note however that there is a direct time tradeoff, that is, the time to change the reference phase shifter through a given phase interval is much longer at 5 m° than it is at 25 m° or 100 m°.

Function 3: LED Display Update Rate

When the function mode is active and the digit “3” appears on the **DISPLAY 1** LED, the period of the display update is shown in seconds on the **DISPLAY 2** LED. It can be changed to any value between 0.1 s and 1.0 s using the **VALUE** ▲ and ▼ keys located below this display.

Function 4: XY Direct On/Off Control

When the function mode is active and the digit “4” appears on the **DISPLAY 1** LED, the status of the XY Direct control is shown on the **DISPLAY 2** LED. It can be turned on or off using the **VALUE** ▲ and ▼ keys located below this display.

The output filters used in the model 5110 normally use analog techniques for time-constants of 30 ms or shorter, with digital methods used at longer settings. When the digital filter is used, the **CH1** and **CH2** outputs are driven by the output of a digital to analog converter.

When the XY Direct mode is turned on, the **CH1** and **CH2** outputs are connected directly to the output filters at all time constant settings. Note however that the mode is automatically turned off when any changes are made to relevant instrument controls, such as sensitivity, time constant, slope, etc.

Function 4: Setting “0”: XY Direct Mode Off

This is the normal setting.

Function 4: Setting “1”: XY Direct Mode On

In this mode the **CH1** and **CH2** outputs are connected directly to the output filters at all time constant settings.

Function 5: Reference Input Mode Control

When the function mode is active and the digit “5” appears on the **DISPLAY 1** LED, the reference input mode is shown on the **DISPLAY 2** LED and can be changed using the **VALUE** ▲ and ▼ keys located below this display. The three settings are as follows:

Function 5: Setting “0”: Internal Reference Mode

In this setting the instrument's reference channel is internally connected to the output of the internal oscillator.

Function 5: Setting “1”: External TTL Reference Mode

In this setting the instrument locks to reference applied to the rear-panel **TTL REF IN** connector.

Function 5: Setting “2”: External Analog Reference Mode

In this setting the instrument locks to reference applied to the front-panel **REF IN** connector.

Function 6: Main ADC Sample Rate

When the function mode is active and the digit “6” appears on the **DISPLAY 1** LED, the main ADC sampling rate is shown in hertz on the **DISPLAY 2** LED and can be changed to any value between 100 Hz and 200 Hz using the **VALUE** ▲ and ▼ keys located below this display.

To leave the function mode press the **+90°/FUNCTION** key once again.

5.5 AUTO Key

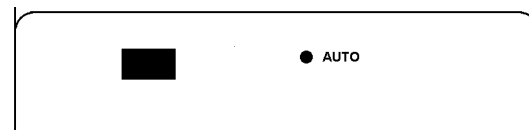


Figure 5-4, AUTO Key

The **AUTO** key, located in its own area beneath the **DISPLAY 1** group and shown above in figure 5-4, is used to execute the various auto-functions. The user invokes an auto-function by pressing the **AUTO** key, which causes the red **AUTO** indicator to light, and then the other key assigned to the function required. In each case, the auto-function is an alternative function for the key and is symbolized in red beneath the key. The red **AUTO** indicator remains lit until execution of the auto-function is completed. An auto-function in progress can be terminated by pressing the **AUTO** key once again.

The auto-functions allow easier, faster operation in most applications. However, given the wide range of possible conditions that could occur, there will be times when performing an operation manually will give better results. During several of the auto-functions, decisions are made based on output readings made at a particular moment. Where this is the case, it is important that the time constant set by the operator be long enough to reduce the output noise to a low level for valid decisions to be obtained. A brief description of each auto-function follows.

5.5.01 Auto-Sensitivity

Secondary function of the **SENSITIVITY** ▲ key. This causes the full-scale sensitivity to be adjusted until the magnitude output lies within the range of 20% to

90% of full-scale.

5.5.02 Auto-Tune

Secondary function of the **TRACK/TUNE** key. If the **MAN** tuning mode is active, pressing the **AUTO** key and then the **TRACK/TUNE** key causes an auto-tune operation to occur, which sets the filter frequency to the reference frequency (in 2F mode to twice the reference frequency). If the tracking mode is selected then auto-tune has no effect.

5.5.03 Auto-Phase

Secondary function of the **DISPLAY 1/PHASE** key. Causes the reference channel phase to be adjusted for maximum X-channel output. During the auto-phase operation, the output time constant assumes preset values which depend on the reference frequency, but it returns to the original setting on completion of the auto-phase cycle.

5.5.04 Auto-Measure

Secondary function of the **SENSITIVITY ▼** key. The auto-measure function provides the fastest, easiest way to take a measurement in many situations. When an auto-measure operation is initiated, the following sequence occurs.

- 1) The signal channel filter is set to bandpass mode, and tuned to the reference frequency.
- 2) The display is set to **% FS/X Y**
- 3) Output offsets and output expand are turned off.
- 4) Normal dynamic reserve is selected.
- 5) The time constant is set to 100 ms and the slope to 12 dB/octave.
- 6) An auto-sensitivity function is performed.
- 7) An auto-phase operation is performed.

The Auto-Measure function is intended to give a quick setting of the instrument which will be approximately correct in typical simple measurement situations. For optimum results in any given situation, it may be convenient to start with Auto-Measure and to make subsequent modifications to the settings of individual controls.

5.5.05 Auto-Offset

Alternative function of the **OFFSET** key. If the **AUTO** key is pressed and then the **OFFSET** key, output offset is applied as required to adjust both the X and Y channel outputs to zero. The offset voltages required to do this can be read on the **DISPLAY 2** LED.

5.6 Analog Panel Meter

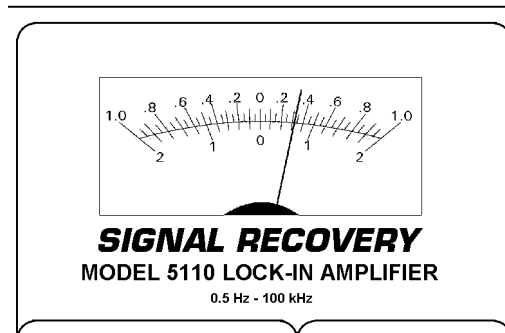


Figure 5-5, Analog Panel Meter

The analog meter, shown above in figure 5-5, always displays the value that is displayed on the **DISPLAY 1** LED when the **DISPLAY 1** selection is set to the **DISP** setting.

5.7 REFERENCE Group

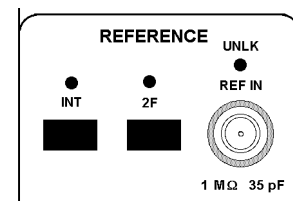


Figure 5-6, REFERENCE Group

The **REFERENCE** group, shown in figure 5-6, contains two keys and associated indicators, a further indicator and the general-purpose external reference input BNC connector, **REF IN**. Another connector, **TTL REF IN**, located on the rear panel, provides an additional reference channel input.

5.7.01 INT Key

The left-most key is used to select external analog or internal reference channel modes, with each press of the key switching between the two states. When the **INT** indicator above the key is extinguished, it is in the external analog reference mode, and the reference channel locks to a signal applied to the front-panel **REF IN** connector. Selection of the rear-panel **TTL REF IN** reference input can be made using the Function 5 control, described earlier in section 5.4.03.

When **INT** is lit, the instrument is in the **INTERNAL** reference mode, and the reference channel locks to the output of the internal oscillator. Where this is the case, the experiment would normally be excited by a signal taken from the **OSC OUT** connector.

5.7.02 2F Key

The second key in this group selects F or 2F operation. When the **2F** indicator is extinguished, F mode is selected and the reference channel operates at the frequency of the applied reference signal or the internal oscillator. When **2F** is lit, 2F mode is selected and the reference channel operates at twice the frequency of the applied reference signal or the internal oscillator.

5.7.03 UNLK indicator

The red **UNLK** (unlock) indicator lights whenever the reference channel loses frequency or phase lock.

5.7.04 REF IN connector

This is the input connector for a general purpose external reference signal. The instrument is fitted with two reference input connectors for external reference mode operation, **TTL REF IN** (rear panel) and **REF IN** (front panel). Wherever possible, use the rear-panel TTL input, particularly if the reference signal is asymmetrical. If the available reference is not suitable for driving a TTL input, then use the front-panel reference input.

The input impedance at this connector is 1 M Ω in parallel with 30 pF.

5.8 OUTPUT Group

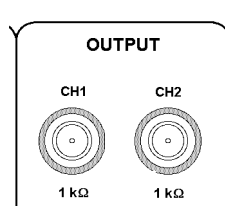


Figure 5-7, OUTPUT Group

The **OUTPUT** group, shown above in figure 5-7, contains the two main analog outputs for the instrument, **CH 1** and **CH 2**. The output signal available at each connector depends on the **DISPLAY 2** selection as discussed further in section 5.9.08.

5.9 DISPLAY 2 Group

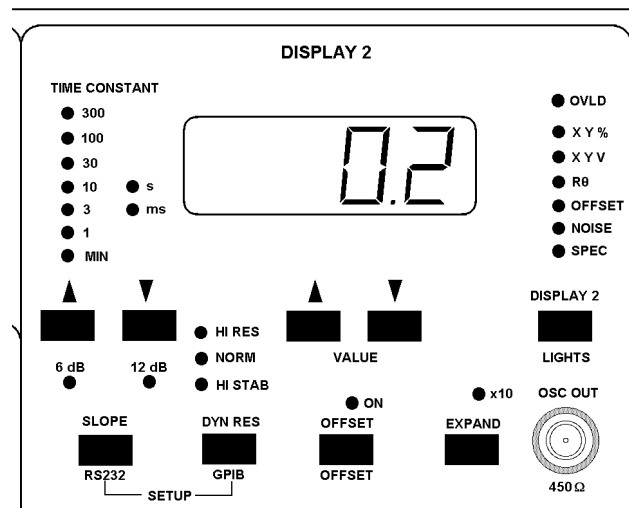


Figure 5-8, DISPLAY 2 Group

The **DISPLAY 2** group, shown above in figure 5-8, contains an LED display and a control to select which of the possible instrument outputs are shown both on it and on the **DISPLAY 1** LED display, when the latter is set to the **DISP** mode. In addition, the group contains controls for the output filter time constant and slope, dynamic reserve, offset, output expansion, and the RS232 and GPIB communications interface settings.

5.9.01 TIME CONSTANT ▲ and ▼ Keys

The primary function of these keys is simply to set the value of the time constant. The longer the time constant, the narrower the lock-in amplifier's noise bandwidth will be and the greater the improvement in signal-to-noise ratio. This improvement is obtained at the expense of increased response time.

Time constants between 1 ms and 300 s may be set in a 1-3-10 sequence, and are applied to both the X- and Y- channel outputs. The present setting is shown by the relevant figure and units indicators being lit. There is also a **MIN** setting at which the nominal time constant is approximately 20 μ s.

5.9.02 SLOPE/RS232 Key

The primary function of this key is to set the output filter's roll-off rate to either 6 dB/octave or 12 dB/octave. 12 dB/octave gives better filtering but cannot be used in some experimental situations (for instance, where the model 5110 is incorporated into a feedback loop). The present setting is indicated by the relevant **6 dB** or **12 dB** indicator being lit.

The key also has a secondary function when used with the green **CONFIG/CNTL** key to select a display of the RS232 interface settings on the **DISPLAY 1** and **DISPLAY 2** LED displays so that these can be changed. This is described later in section 5.10.04

5.9.03 DYN RES/GPIB Key

This is another dual-function key. Its primary function is set the dynamic reserve to high stability, normal or high reserve modes, with the present setting being indicated by the associated **HI STAB**, **NORM** and **HI RES** indicators. The dynamic reserve in **FLAT** filter mode (i.e. with no signal channel filter active) is 20 dB, 40 dB, or 60 dB respectively. Dynamic reserve and output stability are tradeoff parameters, so that the **HI STAB** setting gives an output stability of 5 ppm/°C, **NORM** gives 50 ppm/°C and **HI RES** 500 ppm/°C. The best overall measurement results are therefore obtained with a reserve setting no higher than is necessary to prevent instrument overload. Improving the dynamic reserve by using the input filters does not adversely affect the output stability.

The key also has a secondary function when used with the green **CONFIG/CNTL** key to select a display of the GPIB interface settings on the **DISPLAY 1** and **DISPLAY 2** LED displays so that these can be changed. This is described later in section 5.10.04

5.9.04 OFFSET Key

Again, this is a dual function key. If pressed alone, it simply turns the offset ON or OFF. If an output offset is being applied to the X-channel and Y-channel outputs, then the **ON** indicator above the key is lit. If pressed in conjunction with the **AUTO** key, it initiates an auto-offset cycle.

Offset ON

When the offset **ON** indicator is lit, the value of offset previously set for both X-channel and Y-channel outputs is applied. The offset can be independently set for each output channel when **OFFSET** is selected for display on the **DISPLAY 2** LED. However, it cannot be independently turned on and off for each channel, so that when the offset is on it is on in both channels and when off, it is off in both channels.

Offset off.

When the offset **ON** indicator is unlit, no offset is applied, although any previously-set levels are retained in memory.

Auto-Offset

If the **AUTO** key is pressed and then the **OFFSET** key, output offset is applied as required to adjust both the X and Y channel outputs to zero. The offset voltages required to do this can be read on the **DISPLAY 2** LED, when this is set to the **OFFSET** display mode.

5.9.05 DISPLAY 2 LED ▲ and ▼ keys

The ▲ and ▼ keys below the **DISPLAY 2** LED set the offset value for the selected X-channel or Y-channel output, as follows:-

- 1) Set the **DISPLAY 1** LED to the **DISP** mode.
- 2) Set the **DISPLAY 2** LED to the **OFFSET** mode using the **DISPLAY 2/LIGHTS**

key, discussed later in section 5.9.08.

3) The **DISPLAY 1** LED now shows a digit “1” or “2”, corresponding to the X-channel and Y-channel offset levels respectively. The channel can be changed using the ▲ and ▼ **PARAMETER** keys located below the **DISPLAY 1** LED, whilst the actual offset level for the corresponding channel is shown on the **DISPLAY 2** LED.

4) The offset level for each channel can then be adjusted using the ▲ and ▼ keys below the **DISPLAY 2** LED.

The established values of offset remain in effect until changed.

The keys are also used in conjunction with the green **CONFIG/CNTL** key to set the RS232 and GPIB communications interface settings. This is described later in section 5.10.04

5.9.06 EXPAND Key

The **EXPAND** key toggles the output expansion mode, in which an additional gain of 10 is applied to the X-channel output, after any applied offset. If the mode is on then the **X10** indicator above the key is lit.

5.9.07 OSC OUT connector

This is the output connector for the internal oscillator, which may be set to generate a sine wave between 0.5 Hz and 100 kHz with amplitude between 1 mV and 1.999 V, or 5.000 V rms.

5.9.08 DISPLAY 2/LIGHTS Key

The primary purpose of the **DISPLAY 2/LIGHTS** key is to select the instrument outputs that are displayed on the **DISPLAY 1** and **DISPLAY 2** LED displays and on the analog panel meter. Each press of the key changes the output(s) selected, so that, for example, to change from the % FS/X Y to the X Y mode the key should be pressed once; to return to the % FS/X Y mode it should be pressed a further six times. The present selection is shown by the relevant indicator(s) being lit, and remains in effect until changed.

Note, that in order for output information to be shown on the **DISPLAY 1** LED, it is first necessary that **DISP** (display) be the selected **DISPLAY 1** function.

The choices available are as follows:

% FS Display Mode

When set to the % FS mode, the **DISPLAY 1** LED and the analog panel meter indicate the X-channel output expressed as a percentage of the present full-scale sensitivity. With the reference phase shifter adjusted for maximum X-channel output, a full-scale input signal will give a **DISPLAY 1** LED indication of 100%, a positive full-scale deflection of the analog panel meter and +1 V at the **CH 1** output connector. The **DISPLAY 2** LED indication should be zero and there

should be 0 V at the **CH 2** output connector, indicating the quadrature phase relationship between the reference channel inputs to X-channel and Y-channel PSDs.

% FS/X Y Display Mode

This setting operates in the same way as the **% FS** mode except that, if the sensitivity is set to a “2 range” or a “5 range”, a full-scale input will give an LED indication of “2”, “5”, “20”, “50” “200” or “500”, as opposed to the “100” indication that would occur if **% FS** were selected. This allows the input signal amplitude, in voltage or current units, to be read directly from the LED displays.

For example, if the full-scale sensitivity is set to **20 mV**, the reference phase shifter adjusted for maximum X-channel output and a 15 mV sinusoidal signal applied, then the then **DISPLAY 1** LED indication will be 15. The units applying to this figure are those shown by the lit units indicator in the **SENSITIVITY** section, in this case “**mV**”. Hence the measured signal is 15 mV rms.

In the **% FS/X Y** mode, the analog meter indication and the output levels provided at the **CH 1** and **CH 2** connectors are the same as for **% FS** mode.

% FS/R θ (Magnitude and Phase) Display Mode

In this setting, the magnitude of the signal is shown in terms of a percentage of full-scale on the **DISPLAY 1** LED and on the analog panel meter, and the phase, in degrees, of the signal with respect to the reference, is shown on the **DISPLAY 2** LED. An analog representation of the magnitude output, with a range of 0 to +1 V, is provided at the **CH 1** output connector. Hence if an input signal equal to the present full-scale sensitivity is applied then the voltage at the **CH 1** connector will be +1 V.

The voltage output at the **CH 2** connector is related to the phase shown on the **DISPLAY 2** LED as follows:-

DISPLAY 2 LED	CH2 Voltage
+180	1800 mV
+90	900 mV
0	0 mV
-90	-900 mV
-180	-1800 mV

As can be seen from the table, there is a discontinuity at 180°, that is, the output jumps from +1800 mV to -1800 mV as the phase passes through 180°

R θ (Magnitude and Phase) Display Mode

This mode operates in the same way as the **% FS/R θ** mode except that the magnitude of the signal is shown in directly in volts on the **DISPLAY 1** LED, in the same way as with the **% FS/X Y** mode.

OFFSET Display Mode.

This setting displays the X-channel and Y-channel offset values in effect. The

offset can be set separately for each channel, with the channel selected for offset set/display being shown on the **DISPLAY 1** LED. When this shows the digit "1", the offset value for the X-channel can be read or set using the **DISPLAY 2** LED, whilst if it shows the digit "2", then the offset value for the Y-channel can be read or set using the **DISPLAY 2** LED. Channel selection is accomplished with the ▲ and ▼ **PARAMETER** keys located below the **DISPLAY 1** LED.

The offset value for each channel is set using the ▲ and ▼ keys beneath the **DISPLAY 2** LED, and can be adjusted over the range ± 100.0 , corresponding to $\pm 100\%$ of full-scale. As already mentioned earlier in section 5.9.04, the offset is turned on or off with the **OFFSET/AUTO** key.

NOISE Display Mode

In this setting, the **DISPLAY 2** LCD and the analog meter both indicate the rectified output noise as a percentage of the present full scale sensitivity setting. Because of dynamic range requirements, the quantity actually measured is the mean value of the modulus of X. The result is presented as a percentage of full-scale sensitivity with a scaling factor which gives the correct root-mean-square value if the amplitude distribution is sinusoidal. For the more likely case of a Gaussian distribution, the correct root-mean-square value is obtained by multiplying the reported noise value by a correction factor 1.3313 if the band-pass filter mode is selected, or 1.1986 in the Flat filter mode where a significant contribution to the measured noise is made through windows in the frequency domain centered on harmonics of the reference frequency.

The average is over 1024 samples, taken at a rate equal to the ADC sample rate (Section 5.4.03) and the value of the measured noise output is updated at intervals corresponding to 64 samples.

Note that the settling time of the measured noise output, after a change in the instrument settings or a change in the mean value of X, is $1024/(\text{Sample Rate})$ seconds.

The noise output scaling factor is such that when the X channel output is a steady DC voltage (no noise) the value of the noise output is equal to the value of the X channel output when set to the **% FS** mode. This is a useful way of checking the system calibration when noise measurements are to be made.

Before measuring the output noise, it is normally necessary to null the mean output of the X channel by implementing a suitable phase shift or by use of the X offset facility.

If the purpose of the measurement is to find the spectral density of a broadband random noise source, and assuming that the low-frequency ranges are not in use, the output time constant should typically be set to 10 ms or 1 ms. The band-pass filter option should normally be used so that the spectral density is measured at a well-defined frequency, i.e. the reference frequency; the use of this filter is also important in reducing the effect of interfering signals, but at lower reference frequencies it may have a significant effect on the overall noise bandwidth. When

setting the reference frequency, the power line frequency and its harmonics should be avoided, and a useful precaution is to experiment with small changes in the reference frequency to ensure that the measurement is not being affected by an external interfering signal causing a spurious peak to occur in the measured noise spectrum.

The equivalent noise bandwidth of the output filter is either 1/4TC (6 dB/octave slope) or 1/8TC (12 dB/octave slope) where TC is the selected time-constant.

Because of possible inaccuracies in calculated values of noise bandwidth, the user is strongly recommended to calibrate spectral density measurements by the use of a resistor connected to the input as a noise voltage source. For measurements at frequencies up to a few kHz a value of 100 k Ω is commonly used, giving 40.7 nV/ $\sqrt{\text{Hz}}$ at 300 K. At higher frequencies the input capacitance of the lock-in makes it desirable to use a lower resistor value, but in this case the voltage noise of the lock-in must be taken into account. Note that this method of calibration is appropriate only to those instruments which, like the 5110, have a JFET input stage.

The user is strongly advised to use an oscilloscope attached to the front-panel **SIG MON** (signal monitor) output when making noise measurements as this is the best way of ensuring that one is measuring a random process rather than line pick-up or other non-random signals.

SPEC (Special) Mode

This is the special mode, which provides access to five further output display choices and three controls. When the special display mode is selected, as shown by the **SPEC** indicator being lit, the **DISPLAY 1** LED indicates the number of the selected output or control shown on the **DISPLAY 2** LED, always assuming that **DISP** is selected at **DISPLAY 1**. The **PARAMETER** $\blacktriangle$ and $\blacktriangledown$ keys beneath the **DISPLAY 1** LCD are used to select the required output or control from the following options:

Special Modes 1, 2, 3 and 4

When the **SPEC** mode indicator is lit and the **DISPLAY 1** LED indicates the number 1, 2, 3 or 4, then the **DISPLAY 2** LED shows the voltage present at the corresponding rear-panel **ADC 1** to **ADC 4** connector. The rate at which these inputs are sampled and digitized depends on the ADC trigger mode (see section 5.4.03). The display is in volts, with a range of ± 12 V.

Note that, in any of these four modes, the X-channel output continues to be available at the **CH 1** output connector and is shown on the analog meter. Similarly, the Y-channel output continues to be provided at the **CH 2** output connector.

Special Mode 5 (Ratio Output)

When the **SPEC** mode indicator is lit and the **DISPLAY 1** LED indicates the number 5, the **DISPLAY 2** LED and the analog panel meter indicate the result of the **RATIO** calculation, which is defined as follows:-

$$\text{RATIO} = 10 \times \left(\frac{\text{X output}}{\text{ADC 1 Input}} \right)$$

where X output is the X channel output expressed as a fraction of the full-scale sensitivity and ADC 1 Input is the voltage applied to the **ADC 1** input connector on the rear panel. Hence, for example, if the instrument were measuring a 100 mV signal when set to the 200 mV sensitivity setting, the X channel output were maximized and a 1 V signal were applied to the **ADC 1** input, then the value of RATIO would be:-

$$\text{RATIO} = 10 \times \left(\frac{\frac{0.1}{0.2}}{1.000} \right)$$

$$\text{RATIO} = 0.5$$

The result of the ratio calculation is also presented as an analog voltage at the **CH 1** output connector, and is related to the ratio as follows:

RATIO	CH1 Voltage (volts)
+1	1.0
0	0.0
-1	-1.0

and so in the above example, the **CH 1** output would be 0.5 V. The signal magnitude appears at the **CH 2** output.

Special Modes 6 and 7 (Auxiliary DAC Voltage Control)

When the SPEC mode indicator is lit and the **DISPLAY 1** LED indicates the number 6 or 7, the **DISPLAY 2** LED shows the voltage present at the rear-panel **DAC 1** and **DAC 2** connectors respectively. The voltages can be adjusted using the **▲** and **▼** keys beneath this LCD and can be set to any value between -12.00 V and + 12.00 V in 10 mV increments. The X-channel and Y-channel outputs continue to be provided at the **CH 1** and **CH 2** connectors.

Special Mode 8 (Digital Output Port Control)

When the SPEC mode indicator is lit and the **DISPLAY 1** LED indicates the number 8, the **DISPLAY 2** LED shows a decimal number in the range 0 to 255 representing the status of the eight logic lines provided at the rear-panel **DIGITAL OUTPUTS** connector. The number can be changed using the **▲** and **▼** keys beneath this LCD.

The X-channel and Y-channel outputs continue to be provided at the **CH 1** and **CH 2** connectors.

The **DISPLAY/LIGHTS** key also has a secondary function, when used with the green **CONFIG/CNTL** key, which is to turn the front-panel indicators off and on. This is described later in section 5.10.04

5.9.09 OVLD Indicator

The red **OVLD** indicator in the top right-hand corner of the **DISPLAY 2** group lights when either output channel is overloaded. Output overloads may be removed by one or more of the following actions: reducing the signal-channel sensitivity, inserting the tracking filter or line frequency notch filter, increasing the output time constant, turning off the output expansion function, selecting a higher dynamic range, or establishing a more appropriate value of output offset (including turning it off).

5.10 Interface Group

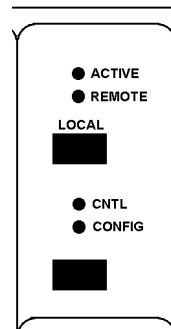


Figure 5-9, Interface Group

The interface group, shown in figure 5-9, includes two keys and four indicators, as follows:

5.10.01 ACTIVE Indicator

This indicator lights when the 5110 is processing commands from either the RS232 or GPIB communications interfaces. If it remains lit, the most likely reason is that a byte is waiting to be read from the GPIB interface.

5.10.02 REMOTE Indicator

This indicator lights when the front-panel controls are locked out of operation by means of the REMOTE computer command. Local operation can be restored by pressing the **LOCAL** key.

5.10.03 LOCAL Key

This key allows the front panel controls to be locked out. Its status is given by the associated **REMOTE** indicator. When the **REMOTE** legend above the key is extinguished, all front-panel controls are operative. When **REMOTE** is lit, the front-panel controls have no effect (with the exception of the **LOCAL** key, which is always operational). The GPIB and RS232 ports are always active.

5.10.04 CONFIG/CNTL Key

This key is tuns the configuration mode on and off, with the present status being shown by the **CONFIG/CNTL** and **CNTL** indicators above the key. When either

indicator is lit, normal instrument operation is suspended, and four other keys allow the communications interface controls to be set and read, the display lights to be turned off or on and the function setting mode to be accessed.

In addition, if the **CONFIG/CNTL** key is pressed twice, the first press causes the GPIB SDC (Selected Device Clear) function to be generated internally. This will clear the buffers, initialize the GPIB Interface and set the value of the Serial Poll Status Byte to "1". The second press restores normal instrument operation.

The alternative functions selected by pressing **CONFIG/CNTL** followed by another key are as follows:

CONFIG/CNTL + SLOPE/RS232C keys

Pressing the **CONFIG/CNTL** key followed by the **SLOPE/RS232C** keys allows the RS232 communications settings to be set or read. In this mode, the **DISPLAY 1** LED shows a control code, either "0" or "1", whilst the **DISPLAY 2** LED shows a value code corresponding to the control code. The control code can be changed using the **PARAMETER ▲** and **▼** keys beneath the **DISPLAY 1** LED, and the value code can be adjusted using the **VALUE ▲** and **▼** keys below the **DISPLAY 2** LED.

The settings are as follows:-

RS232 Baud Rate Control: Display 1 LED = "0"

When the **DISPLAY 1** LED shows a "0", the **DISPLAY 2** LED shows the baud rate value code. To change the baud rate (range 110 baud to 19.2 kilobaud), use the **VALUE ▲** and **▼** keys below the **DISPLAY 2** LED to obtain the required value code. The baud rates corresponding to each value code are as follows:

Value Code	Baud Rate
0	75
1	110
2	134.5
3	150
4	300
5	600
6	1200
7	2000
8	2400
9	4800
10	1800
11	9600
12	19200

RS232 Other Parameters Control: Display 1 LED = "1"

When the **DISPLAY 1** LED shows a "1", the **DISPLAY 2** LED shows a value code which determines the setting of the other six RS232 interface controls, i.e. Number of Data Bits, Parity Odd or Even, Parity On or Off, Number of Stop Bits, Character Echo status and Prompt Character status. To change these controls, use the **VALUE ▲** and **▼** keys below the **DISPLAY 2** LED to obtain the required

value code.

The value code is a decimal number between 0 and 63 given by the sum of five numbers, as follows:

Control Setting	Value
Seven Data Bits	0
Eight Data Bits	1
Even Parity	0
Odd Parity	2
One Stop Bits	0
Two Stop Bit	4
Parity ON	0
Parity OFF	8
RS232 Echo OFF	0
RS232 Echo ON	16
RS232 Prompt Enabled	0
RS232 Prompt Disabled	32

For example, a value code of 16 would indicate:

DATA BITS: 7
 PARITY: ON and EVEN
 STOP BITS: 1
 RS232 ECHO: ON
 RS232 PROMPT: ENABLED

Once the RS232 interface controls are set as required, press the **CONFIG/CNTL** key again to restore normal instrument operation.

CONFIG/CNTL + DYN RES/GPIB keys

Pressing the **CONFIG/CNTL** key followed by the **DYN RES/GPIB** keys allows the GPIB communications settings to be set or read. In this mode, the **DISPLAY 1** LED shows a control code, either “0” or “1”, whilst the **DISPLAY 2** LED shows a value code corresponding to the control code. The control code can be changed using the **PARAMETER ▲** and **▼** keys beneath the **DISPLAY 1** LED, and the value code can be adjusted using the **VALUE ▲** and **▼** keys below the **DISPLAY 2** LED.

GPIB Address Control: Display 1 LED = “0”

When the **DISPLAY 1** LCD shows a “0”, the **DISPLAY 2** LCD shows the GPIB address. To change the address (range 0 to 31), use the **VALUE ▲** and **▼** keys below the **DISPLAY 2** LCD.

GPIO Terminator and GPIO Test Echo Controls:**Display 1 LED = "1"**

When the **DISPLAY 1** LED shows a "1", the **DISPLAY 2** LED shows a value code which sets the termination and test echo controls. To change these controls, use the **VALUE** ▲ and ▼ keys below the **DISPLAY 2** LED to obtain the required value code.

The value code is a decimal number between 0 and 5, as follows:-

Value Code	Function
0	<CR>, Test Echo OFF
1	<CR>, Test Echo ON
2	<CR,LF>, Test Echo OFF
3	<CR,LF>, Test Echo ON
4	No terminator, Test Echo OFF
5	No terminator, Test Echo ON

where <CR> indicates the Carriage Return (ASCII 13) and <LF> the Line Feed (ASCII 10) terminator characters.

The 5110 accepts as GPIO input terminator whichever terminator option is selected. A controlling computer, on the other hand, will probably operate with only one terminator or the other, making it essential that the correct terminator be selected. Frequently, the easiest way to determine which terminator the computer requires is to simply try them both and see which one works.

When the test echo function is turned on, every character transmitted or received via the GPIO interface is echoed to the RS232 interface. This mode can be useful when troubleshooting GPIO communications problems.

NOTE: Test Echo should be turned off when it is not in use since it slows down GPIO communications.

Pressing the **CONFIG/CNTL** key again restores normal operation.

CONFIG/CNTL + DISPLAY/LIGHTS keys

The **CONFIG/CNTL** key is also used with the **DISPLAY/LIGHTS** key to turn the front-panel indicators off and on. Operation with the lights turned off is often useful in those optical experiments where stray light must be minimized.

To turn the lights off, first press the **CONFIG/CNTL** key and then the **DISPLAY/LIGHTS** key. All lights will immediately extinguish but the 5110 will not operate correctly. When **CONFIG/CNTL** is pressed a second time, normal lock-in amplifier operation (with all panel lights extinguished) will be established.

To turn the lights on, first press **CONFIG/CNTL** key and then the **DISPLAY/LIGHTS** key. The panel lights will become operational but the lock-in amplifier will not operate correctly. When **CONFIG/CNTL** is pressed a second time, normal lights-on operation of the lock-in amplifier will return.

CONFIG/CNTL + +90°/FUNCTION keys

When the **CONFIG/CNTL** and **+90°/FUNCTION** keys are pressed in succession, the **CNTL** indicator lights and the instrument enters the function mode, which gives access to controls, such as ADC sample rate, which rarely need to be adjusted. These controls have already been fully described in section 5.4.03

5.11 POWER on/off key

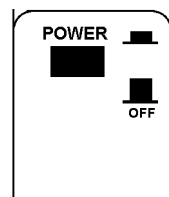


Figure 5-10, POWER on/off key

The **POWER** on/off key, shown in figure 5-10, controls the line power to the instrument. In the latched (in) position, the power is on. In the released (out) position, it interrupts both sides of the AC power to the primary of the power transformer.

5.12 Typical Lock-in Amplifier Experiment

The model 5110 is a complex instrument with many controls and so readers may find the following basic checklist helpful in setting up the instrument for manual operation.

Selection of Signal Input

Use the upper four right-hand interlocking keys in the **SENSITIVITY** group to select voltage (single ended or differential) or current input mode, and connect the signal source to the relevant **A** and/or **B/I VIR GND** input connector(s).

Selection of Reference Mode

Use the reference source selection control in the **REFERENCE** group to set the instrument to internal reference mode, assuming that the internal oscillator will be used as a source of excitation to your experiment. Use the **SET OSC LVL** oscillator amplitude and **SET OSC F** frequency controls in the **DISPLAY 1** group to set the required oscillator output, and connect the output signal from the **OSC OUT** connector to the experiment.

If using external reference mode, use the reference source selection control in the **REFERENCE** group to set the instrument to external reference mode, and connect the reference signal to the front-panel **REF IN** connector.

Auto-Measure

Press the red **AUTO** key followed by the **SENSITIVITY ▼** key to perform an auto-measure operation, which will set the instrument so that it is correctly measuring the signal.

Other Adjustments

Use the **DISPLAY 2** selection key to set the two LED displays so that they are displaying the required instrument outputs.

6.1 Introduction

The model 5110 includes both RS232 and GPIB (IEEE488) interface ports, designed to allow the lock-in amplifier to be completely controlled from a remote computer. Virtually all of the instrument's controls may be operated, and all the outputs read, via these interfaces. In addition, there are a few functions, such as setting the internal oscillator output amplitude to +5.000 V rms and changing the frequency setting of the line frequency rejection filter which can only be accessed remotely.

This chapter describes the capabilities of the instrument when operated remotely and discusses how this is done.

6.2 Capabilities

6.2.01 General

All instrument controls which may be set using the front panel keys may also be set remotely, with the exception of the selection of voltage or current input mode, the current mode converter settings and the setting of the input BNC connector's float/ground status. In addition, the present setting of each control, with the exception of the same features, may be determined by the computer. All instrument outputs which may be displayed on the front panel may also be read remotely.

When operated via the interfaces, the following features are also available:

6.2.02 Oscillator amplitude - additional output settings

The front panel **OSC LVL** control allows the internal oscillator amplitude to be set to voltages between 1 mV and 1.999 mV rms. However, when setting this voltage by computer command, it may also be set to 2.000 V and 5.000 V rms.

6.2.03 Line Filter Frequency

The line frequency rejection filter is normally set to suit the line frequency of the country of destination. However if the instrument is moved from a 60 Hz to 50 Hz region or vice-versa, then the filter setting can be changed by the use of a special command.

6.3 RS232 and GPIB Operation

6.3.01 Introduction

Control of the lock-in amplifier from a computer is accomplished by means of communications over the RS232 or GPIB interfaces. The communication activity consists of the computer sending commands to the lock-in amplifier, and the lock-in amplifier responding, either by sending back some data or by changing the setting of

one of its controls. The commands and responses are encoded in standard 7-bit ASCII format, with one or more additional bits as required by the interface (see below).

The two ports cannot be used simultaneously, but when a command has been completed, the lock-in amplifier will accept a command at either port and send any response to the same port. Also when the test echo facility has been activated all output from the computer to the GPIB can be monitored by a terminal attached to the RS232 connector.

Although the interface is primarily intended to enable the lock-in amplifier to be operated by a computer program specially written for an application, it can also be used in the direct, or terminal, mode. In this mode the user enters commands on a keyboard and reads the results on a video screen.

The simplest way to establish the terminal mode is to connect a standard terminal, or a terminal emulator, to the RS232 port. A terminal emulator is a computer running special purpose software that makes it act as a terminal. In the default (power-up) state of the port, the lock-in amplifier sends a convenient prompt character when it is ready to receive a command, and echoes each character that is received.

Microsoft Windows versions 3.1 and 3.11 include a program called Terminal, and Windows 95 a program called HyperTerminal, usually in the Accessories group, which may be used as a terminal emulator. On the other hand, a simple terminal program with minimal facilities can be written in a few lines of BASIC code (see appendix C.1).

6.3.02 RS232 Interface - General Features

The RS232 interface in the model 5110 is implemented with three wires; one carries digital transmissions from the computer to the lock-in amplifier, the second carries digital transmissions from the lock-in amplifier to the computer and the third is the logic ground to which both signals are referred. The logic levels are $\pm 12\text{V}$ referred to logic ground, and the connection may be a standard RS232 cable in conjunction with a null modem or alternatively may be made up from low-cost general purpose cable. The pinout of the RS232 connector is shown in appendix B and cable diagrams suitable for coupling the instrument to a computer are shown in appendix D.

The main advantages of the RS232 interface are:

- 1) It communicates via a serial port which is present as standard equipment on nearly all computers, using leads and connectors which are available from suppliers of computer accessories or can be constructed at minimal cost in the user's workshop.
- 2) It requires no more software support than is normally supplied with the computer, for example Microsoft's GWBASIC, QBASIC or Windows Terminal or Hyperterminal mode.

A single RS232 transmission consists of a start bit followed by 7 or 8 data bits, an optional parity bit, and 1 stop bit. The rate of data transfer depends on the number of

bits per second sent over the interface, usually called the baud rate. In the model 5110 the baud rate can be set to a range of different values up to 19,200, corresponding to a minimum time of less than 0.5 ms for a single character.

Mostly for historical reasons, there are a very large number of different ways in which RS232 communications can be implemented. Apart from the baud rate options, there are choices of data word length (7 or 8 bits), parity check operation (even, odd or none), and number of stop bits (1 or 2). These settings may be adjusted using the **CONFIG/CNTL** and **SLOPE/RS232** keys, as described in section 5.10.04. They may also be adjusted by means of the RS command.

NOTE: In order to achieve satisfactory operation, the RS232 settings must be set to exactly the same values in the terminal or computer as in the lock-in amplifier.

6.3.03 Choice of Baud Rate

Where the lock-in amplifier is connected to a terminal or to a computer implementing an echo handshake, the highest available baud rate of 19,200 is normally used if, as is usually the case, this rate is supported by the terminal or computer. Lower baud rates may be used in order to achieve compatibility with older equipment or where there is some special reason for reducing the communication rate.

6.3.04 Choice of Number of Data Bits

The model 5110 lock-in amplifier uses the standard ASCII character set, containing 127 characters represented by 7-bit binary words. If an 8-bit data word is selected, the most significant bit is set to zero on output from the lock-in amplifier and ignored on input. The result is that either the 8-bit or the 7-bit option may be used, but the 7-bit option gives slightly faster communication.

6.3.05 Choice of Parity Check Option

Parity checks are not required at the baud rates available in the model 5110, that is up to 19,200 baud, with typical cable lengths of up to a few meters. Therefore no software is provided in the model 5110 for dealing with parity errors. Where long cables are in use, it may be advisable to make use of a lower baud rate. The result is that any of the parity check options may be used, but the parity off option will result in slightly faster communication.

Where the RS232 parameters of the terminal or computer are capable of being set to any desired value, an arbitrary choice must be made. In the model 5110 the combination set at the factory is even parity check, 7 data bits, and one stop bit because these are the MSDOS default.

6.3.06 GPIB Interface General Features

The GPIB is a parallel digital interface with 8 bi-directional data lines, and 8 further lines which implement additional control and communication functions. Communication is through 24-wire cables (including 8 ground connections) with special-purpose connectors which are constructed in such a way that they can be stacked on top of one another to enable numerous instruments to be connected in

parallel. By means of internal hardware or software switches, each instrument is set to a different address on the bus, usually a number in the range 0 to 31. In the model 5110 the address is set using the **CONFIG/CNTL** and **DYN RES/GPIB** keys, discussed in section 5.10.04 or by means of the GP command.

A most important aspect of the GPIB is that its operation is defined in minute detail by the IEEE488 standard, usually implemented by special purpose semiconductor devices that are present in each instrument and communicate with the instrument's microprocessor. The existence of this standard greatly simplifies the problem of programming the bus controller, i.e. the computer, to implement complex measurement and test systems involving the interaction of numerous instruments. There are fewer interface parameters to be set in the lock-in amplifier than with RS232 communications.

The operation of the GPIB requires the computer to be equipped with special-purpose hardware, usually in the form of a plug-in card, and associated software which enable it to act as a bus controller. The control program is written in a high-level language, usually BASIC or C, containing additional subroutines implemented by software supplied by the manufacturer of the interface card.

Because of the parallel nature of the GPIB and its very effective use of the control lines including the implementation of a three-wire handshake (see below), comparatively high data rates are possible, up to a few hundred thousand bytes per second. In typical setups the data rate of the GPIB itself is not the factor that limits the rate of operation of the control program.

6.3.07 Handshaking and Echoes

A handshake is a method of ensuring that the transmitter does not send a byte until the receiver is ready to receive it, and, in the case of a parallel interface, that the receiver reads the data lines only when they contain a valid byte.

GPIB Handshaking

The GPIB interface includes three lines (*DAV, *NRFD, *NDAC) which are used to implement a three-wire handshake. The operation of this is completely defined by the IEEE488 standard and is fully automatic, so that the user does not need to know anything about the handshake when writing programs for the GPIB. Note that each command must be correctly terminated.

RS232 Handshaking

In the RS232 standard there are several control lines called handshake lines (RTS, DTR outputs and CTS, DSR, DCD inputs) in addition to the data lines (TD output and RD input). However, these lines are not capable of implementing the handshaking function required by the model 5110 on a byte-by-byte basis and are not connected in the model 5110 apart from the RTS (request to send) and DTR (data terminal ready) inputs, which are permanently asserted.

Note that some computer applications require one or more of the computer's RS232 handshake lines to be asserted. If this is the case, and if the requirement cannot be changed by the use of a software switch, the cable may be used in conjunction with a

null modem. A null modem is an adapter which connects TD on each side through to RD on the other side, and asserts CTS, DSR, and DCD on each side when RTS and DTR are asserted on the other side.

With most modern software there is no need to use any of the RS232 handshake lines and a simple three-wire connection can be used. The actual handshake function is then performed by means of bytes transmitted over the interface.

The more critical handshake is the one controlling the transfer of a command from the computer to the lock-in amplifier, because the computer typically operates much faster than the lock-in amplifier and bytes can easily be lost if the command is sent from a program. (Note that because of the limited speed of human typing, there is no problem in the terminal mode.) Therefore an *echo handshake* is used, which works in the following way: after receiving each byte, the lock-in amplifier sends back an echo, that is a byte which is a copy of the one that it has just received, to indicate that it is ready to receive the next byte. Correspondingly, the computer does not send the next byte until it has read the echo of the previous one. Usually the computer makes a comparison of each byte with its echo, and this constitutes a useful check on the validity of the communications.

Where the echo is not required, it can be suppressed by negating bit 3 in the RS232 parameter byte. The default (power-up) state of this bit is for it to be asserted.

The program RSCOM2.BAS in section C.2 illustrates the use of the echo handshake.

In some cases, timing errors may occur in the communication of the lock-in amplifier's responses to the computer. The 5110 provides for this with the *data request mode* entered by means of the DRQMODE command. In this mode the lock-in amplifier will not send any response byte over the RS232 interface unless the computer has previously sent the data request byte, which is a pound (hash) symbol, # (ASCII character 35). Each data request byte sent by the computer causes one response byte to be sent by the lock-in amplifier.

6.3.08 Terminators

In order for communications to be successfully established between the lock-in amplifier and the computer, it is essential that each transmission, i.e. command or command response, is terminated in a way which is recognizable by the computer and the lock-in amplifier as signifying the end of that transmission.

In the model 5110 there are three input termination options for GPIB communications, selected from the front panel using the **CONFIG/CNTL** and **DYN RES/GPIB** keys or by means of the GP command. The lock-in amplifier may be set to expect the <CR> byte (ASCII 13), the <CR,LF> sequence (ASCII 13 followed by ASCII 10) or the GPIB line EIO (End or Identify) to be appended by the controller as a terminator to the end of each command.

The selected GPIB termination option applies also to the output termination of any responses sent back by the lock-in amplifier to the controller, i.e. the lock-in amplifier will send <CR> or <CR,LF> as appropriate. In all cases the lock-in amplifier asserts

the GPIB EOI (end or identify) signal line during the transmission of the last byte of a response.

In RS232 communications, the lock-in amplifier automatically accepts either <CR> or <CR,LF> as an input command terminator, and sends out <CR,LF> as an output response terminator.

6.3.09 Command Format

The simple commands listed in section 6.4 have one of five forms:

CMDNAME terminator
CMDNAME n terminator
CMDNAME [n] terminator
CMDNAME [n₁ [n₂]] terminator
CMDNAME n₁ [n₂] terminator

where CMDNAME is an alphanumeric string that defines the command, and n, n₁, n₂ are parameters separated by spaces. When n is not enclosed in square brackets it must be supplied. [n] means that n is optional. [n₁ [n₂]] means that n₁ is optional and if present may optionally be followed by n₂. Uppercase and lowercase characters are equivalent. Terminator bytes are defined in section 6.3.08.

NOTE: Where the command syntax includes optional parameters and the command is sent without the optional parameters, the response consists of a transmission of the present values of the parameter(s).

Any response transmission consists of one or more numbers followed by a response terminator. Where the response consists of two or more numbers in succession, they are separated by a delimiter.

6.3.10 Delimiters

Any response transmissions consist of one or two numbers followed by a response terminator. Where the response of the lock-in amplifier consists of two numbers in succession, they are separated by a byte called a delimiter. This delimiter can be any printing ASCII character and is selected by the DD command.

6.3.11 Compound Commands

A compound command consists of two or more simple commands separated by semicolons (ASCII 59) and terminated by a single command terminator. If any of the responses involve data transmissions, each one is followed by an output terminator.

6.3.12 Status Byte, Prompts and Overload Byte

An important feature of the GPIB standard is the serial poll operation by which a special byte, the status byte, may be read at any time from any instrument on the bus. This contains information which must be urgently conveyed from the instrument to the controller.

The function of the individual bits in the status byte is instrument dependent, apart from bit 6 (the request service bit) whose function is defined by the standard.

In the model 5110, bits 0 and 7 signify "command complete" and "data available" respectively. In GPIB communications, the use of these bits can lead to a useful simplification of the control program by providing a response subroutine which is the same for all commands, whether or not they send a response over the bus. The subroutine should carry out the following sequence of events:

- 1) Send the command
- 2) Perform repeated serial poll operations in a loop until bit 0 (command complete) is zero. This implies that the command has been received by the instrument and is being processed. However, it is possible that if the command were a write-only command and a slow computer were being used then the instrument may actually clear and then reset bit 0 (i.e. actually complete the command) before the first serial poll operation were executed. Hence the loop must include the provision to timeout under these conditions; a value of 10 ms should be satisfactory in most cases.
- 3) If bit 0 clears, perform repeated serial polls testing both bit 0 and bit 7 (data available) and, if bit 7 is asserted then perform a read operation. This cycle (i.e. test bit 0 (command complete) and test bit 7 (data available)) should then continue until the lock-in amplifier asserts bit 0 to indicate that the command-response sequence is complete, so that the instrument will then be ready for the next command.

This procedure, although apparently complex, deals successfully with compound commands.

In RS232 communications, comparatively rapid access to the status byte is provided by the prompt character which is sent by the lock-in amplifier at the same time as bit 0 becomes asserted in the status byte. This character is sent out by the lock-in amplifier after each command response (whether or not the response includes a transmission over the interface) to indicate that the response is finished and the instrument is ready for a new command. The prompt takes one of two forms. If the command contained an error, either in syntax or by a command parameter being out of range, or alternatively if an overload or reference unlock is currently being reported by the front panel indicators, the prompt is ? (ASCII 63). Otherwise the prompt is * (ASCII 42).

These error conditions correspond to the assertion of bits 1, 2, 3 or 4 in the status byte. When the ? prompt is received by the computer, the ST command may be issued in order to discover which type of fault exists and to take appropriate action.

If the RS232 prompt character is not required it can be disabled by sending the RS command with bit 5 in the second parameter asserted.

The prompts are a rapid way of checking on the instrument status and enable a convenient keyboard control system to be set up simply by attaching a standard terminal, or a simple computer-based terminal emulator, to the RS232 port.

Because of the limited number of bits in the status byte, it can indicate that an overload exists but cannot give more detail. An auxiliary byte, the overload byte returned by the N command, gives details of the location of the overload. A summary of the bit assignments in the status byte and the overload byte is given below.

Bit	Status Byte	Overload Byte
0	command complete	not used
1	invalid command	not used
2	command parameter error	not used
3	reference unlock	Y channel output overload
4	overload	X channel output overload
5	auto function in progress	PSD overload
6	SRQ asserted	input overload
7	data available	reference unlock

Note that bit 5 is set while an auto-function is in progress.

6.3.13 Service Requests

The GPIB interface includes a line (pin 10 on the connector) called the SRQ (service request) line which is used by the instrument to signal to the controller that urgent attention is required. At the same time that the instrument asserts the SRQ line, it also asserts bit 6 in the status byte. The controller responds by executing a serial poll of all the instruments on the bus in turn and testing bit 6 of the status byte in order to discover which instrument was responsible for asserting the SRQ line. The status byte of that instrument is then further tested in order to discover the reason for the service request and to take appropriate action.

In the model 5110 the assertion of the SRQ line is under the control of a byte called the SRQ mask byte which can be set by the user with the MSK command. If any bit in the status byte becomes asserted, and the corresponding bit in the mask byte has a nonzero value, the SRQ line is automatically asserted. If the value of the mask byte is zero, the SRQ line is never asserted.

Hence, for example, if the SRQ mask byte is set to 16, a service request would be generated as soon as an overload occurred; if the SRQ mask byte were set to 0, then service requests would never be generated.

6.4 Command Descriptions

This section lists the commands in logical groups, so that, for example, all commands associated with setting controls affecting the signal channel are shown together. Appendix E gives the same list of commands but in alphabetical order.

6.4.01 Signal Channel

SEN [n] Full-scale sensitivity control

The value of parameter n sets or reads the full-scale sensitivity according to the following table:

n	full-scale sensitivity
0	100 nV
1	200 nV
2	500 nV
3	1 μ V
4	2 μ V
5	5 μ V
6	10 μ V
7	20 μ V
8	50 μ V
9	100 μ V
10	200 μ V
11	500 μ V
12	1 mV
13	2 mV
14	5 mV
15	10 mV
16	20 mV
17	50 mV
18	100 mV
19	200 mV
20	500 mV
21	1 V

AS Perform an Auto-Sensitivity operation

The instrument adjusts its full-scale sensitivity so that the magnitude output lies between 20 % and 90 % of full-scale.

ASM Perform an Auto-Measure operation

The instrument adjusts its full-scale sensitivity so that the magnitude output lies between 20 % and 90 % of full-scale, and then performs an auto-phase operation to maximize the X-channel output and minimize the Y-channel output.

6.4.02 Signal Channel Filters

FLT [n] Main filter operating mode

The value of parameter n sets or reads the signal-channel filter mode according to the following table:

n	mode
0	FLAT (no filter)
1	LP (low-pass filter)
2	BP (band-pass filter)

ATC [n] Filter frequency tuning mode control
The value of parameter n sets or reads the filter frequency tuning mode according to the following table:

- n filter frequency tuning mode
- 0 manual (frequency set by front-panel keys or by the FF command)
- 1 automatic (filter frequency set to reference channel frequency)

FF [n₁, n₂] Filter frequency control
The value of parameter n₂ sets or reads the filter frequency band according to the following table:

- n₂ frequency band
- 0 0.5 Hz to 12 Hz
- 1 10 Hz to 120 Hz
- 2 100 Hz to 1200 Hz
- 3 1 kHz to 12 kHz
- 4 10 kHz to 105 kHz

The value of parameter n₁ sets or reads the frequency within each band in the range 1000 to 12000, except in the bottom band (n₂ = 0) where the range is 500 to 12000 and in the top band (n₂ = 4) where the range is 1000 to 10500

Hence, for example, to set the frequency to 1 kHz, send the commands FF 1000 3 or FF 10000 2

ATS Auto-tune filter frequency
The ATS command causes the filter frequency to be tuned to the reference frequency, by setting the filter track mode on and then off.

LF [n] Signal channel line frequency rejection filter control
The value of parameter n sets or reads the mode of the line frequency notch filter according to the following table:

- n selection
- 0 Off
- 1 Enable 50 or 60 Hz notch filter
- 2 Enable 100 or 120 Hz notch filter
- 3 Enable both filters

LINE [n] Line frequency rejection filter center frequency control
The value of parameter n sets or reads the center frequencies of the line frequency notch filter according to the following table:

- n selection
- 0 50 and/or 100 Hz
- 1 60 and/or 120 Hz

6.4.03 Reference Channel

IE [n] Reference channel source control (Internal/External)

The value of parameter n sets or reads the reference input mode according to the following table:

n	selection
0	INT (Internal)
1	EXT TTL (External rear-panel TTL REF IN input)
2	EXT Analog (External front panel REF IN input)

LOCK Lock reference channel

Causes a reference lock operation, updating all frequency-dependent calibration parameters.

F2F [n] Reference harmonic mode control

The value of parameter n sets or reads the reference frequency mode, according to the following table:

n	selection
0	Signal detection occurs at the applied reference frequency (internal or external)
1	Signal detection occurs at twice the applied reference frequency (internal or external)

P [n₁, n₂] Reference phase control

The parameters n₁ and n₂ set or read the reference channel phase shifter, as follows:-

Parameter n₁ sets or reads the phase quadrant according to the following table:

n ₁	quadrant
0	0° to 90°
1	90° to 180°
2	180° to 270°
3	270° to 360°

Parameter n₂ sets or reads the phase within each quadrant to any value between 0° and 100° in millidegrees, with a minimum resolution of 5 m°, i.e. n₂ can range from 0 to 100000.

AQN Auto-Phase (auto quadrature null)

The instrument adjusts the reference phase to maximize the X channel output and minimize the Y channel output signals.

FRQ Reference frequency meter

The FRQ command causes the lock-in amplifier to respond with 0 if the reference channel is unlocked, or with the measured reference frequency expressed in millihertz if it is locked.

6.4.04 Signal Channel Output Filters

XDB [n] Output low-pass filter slope (roll-off) control

The value of parameter n sets or reads the slope of the output filters according to the following table:

n	slope
0	6dB/octave
1	12dB/octave

XTC [n] or

TC [n] Filter time constant control

The value of parameter n sets or reads the time constant of the output according to the following table:

n	time constant
0	MIN
1	1 ms
2	3 ms
3	10 ms
4	30 ms
5	100 ms
6	300 ms
7	1 s
8	3 s
9	10 s
10	30 s
11	100 s
12	300 s

6.4.05 Signal Channel Output Amplifiers

DR [n] Dynamic reserve control

The value of parameter n sets or reads the instrument's dynamic reserve, according to the following table:

n	selection
0	HI STAB (high stability setting)
1	NORM (normal setting)
2	HI RES (high dynamic reserve setting)

XOF [n₁ [n₂]] X channel output offset control

The value of parameter n₁ sets or reads the status and level of the X offset facility according to the following table:

n ₁	selection
0	Disables X offset facility
1	Enables X offset facility

The range of n₂ is ± 1000 corresponding to $\pm 100\%$ of the full-scale output.

YOF [n_1 [n_2]] Y channel output offset control

The value of parameter n_1 sets or reads the status and level of the Y offset facility according to the following table:

n_1	selection
0	Disables Y offset facility
1	Enables Y offset facility

The range of n_2 is ± 1000 corresponding to ± 100 % of the full-scale output.

AXO Auto-Offset

The X and Y channel output offsets are turned on and set to levels giving zero X and Y channel outputs. Any changes in the input signal then appear as changes about zero in the outputs.

EX [n] Output expansion control

The value of parameter n sets or reads the X-channel output expansion mode in accordance with the following table:

n	expand mode
0	Off
1	Expand X-channel output by $\times 10$.

6.4.06 Instrument Outputs

X X channel output

The X command causes the lock-in amplifier to respond with the X-channel output in the range ± 12000 , full-scale being ± 10000

Y Y channel output

The Y command causes the lock-in amplifier to respond with the Y-channel output in the range ± 12000 , full-scale being ± 10000

XY X, Y channel outputs

Equivalent to the compound command X;Y

MAG Magnitude

The MAG command causes the lock-in amplifier to respond with the magnitude output in the range 0 to $+12000$, full-scale being $+10000$

PHA Signal phase

The PHA command causes the lock-in amplifier to respond with the signal phase in millidegrees, in the range ± 180000 , corresponding to $\pm 180^\circ$.

MP Magnitude, phase

Equivalent to the compound command MAG;PHA

RT Ratio output

The RT command reports a number equivalent to $(1000 \times X)/\text{ADC1}$ where X is the

value that would be returned by the X command and ADC1 is the value that would be returned by the ADC1 command.

NN Noise output
The NN command causes the lock-in amplifier to respond with the noise, expressed as a percentage of the present full scale sensitivity, with 10000 corresponding to 100%. The noise is measured in a bandwidth defined by the output filter time constant and slope, centered about the reference frequency.

6.4.07 Internal Oscillator

OA [n] Oscillator amplitude control
The value of parameter n sets or reads the oscillator amplitude. The range of n is 0 to 2000 representing 0 to 2 V rms, but the value 5000 may also be used, which corresponds to an output amplitude of 5.000 V rms.

OF [n₁ n₂] Oscillator frequency control
The value of parameter n₂ sets or reads the oscillator frequency band according to the following table:

n ₂	frequency band
0	0.5 Hz to 2 Hz
1	1.8 Hz to 20 Hz
2	18 Hz to 200 Hz
3	180 Hz to 2 kHz
4	1.8 kHz to 20 kHz
5	18 kHz to 105 kHz

The value of parameter n₁ sets or reads the frequency within each band in the range 1800 to 20000, except for the bottom band (n₂ = 0) where the range is 5000 to 20000 and the top band (n₂ = 5) where the range is 1800 to 10500.

Hence, for example, to set the frequency to 1 kHz, send the command OF 10000 3.

6.4.08 Auxiliary Outputs

DAC1 [n] Auxiliary DAC 1 output control
The value of parameter n sets or reads the voltage appearing at the rear panel **DAC 1** output, in the range -12000 to +12000, corresponding to voltages from -12.000 V to +12.000 V.

DAC2 [n] Auxiliary DAC 2 output control
The value of parameter n sets or reads the voltage appearing at the rear panel **DAC 2** output, in the range -12000 to +12000, corresponding to voltages from -12.000 V to +12.000 V.

BYTE [n] Digital output port control
The value of n, in the range 0 to 255, determines the bits to be output on the rear panel digital output port. When n = 0, all outputs are low, and when n = 255, all are high.

6.4.09 Auxiliary Inputs

ADC n Read auxiliary analog to digital inputs

Reads the voltage appearing at the rear panel **ADC 1** ($n = 1$) or , **2** ($n = 2$), **3** ($n = 3$) or **4** ($n = 4$) inputs.

The response for ADC n is an integer in the range -12000 to +12000, corresponding to voltages from -12.000 V to +12.000 V.

TRIG [n] ADC trigger mode control

The value of parameter n sets or reads the trigger mode of the ADC converter used for converting both the X-channel and Y-channel outputs and the auxiliary ADC inputs according to the following table:

n trigger mode

0 Reference Trigger Mode

In this setting, the ADC is triggered synchronously with the reference frequency. The maximum conversion rate is nominally once every 5 ms (200 Hz), so that at low reference frequencies there will be one trigger per reference period. At higher reference frequencies there will be one trigger per some multiple of the reference period.

Each trigger results in the conversion of X- and Y-channel outputs and one of the four auxiliary **ADC** inputs. Thus any given auxiliary **ADC** input is converted once every four trigger cycles.

1 External Trigger Mode

In this setting the ADC is triggered once on each positive going TTL transition detected at rear-panel **ADC EXT TRIG IN** connector. If an applied trigger occurs before the converter has finished processing the previous trigger, it is ignored. On any given trigger, the X- and Y-channel outputs and all four of the auxiliary **ADC** inputs are converted.

2 Asynchronous Trigger Mode

In this setting the ADC is triggered asynchronously at the preset system sample rate, which is adjustable over the range 100 Hz to 200 Hz by means of a front panel control or the **SAMPLE** computer command. On any given trigger, the X- and Y-channel outputs and one of the four auxiliary **ADC** inputs are converted. Thus any given auxiliary **ADC** input is converted once every four trigger cycles (i.e. at a quarter of the system sample rate).

4 Ratio Trigger Mode

In this setting, triggering is synchronous with the reference signal as for mode 0, but only the X- and Y- channel outputs and the **ADC 1** input are converted. The maximum conversion rate is nominally once every 5 ms (200 Hz), so that at low reference frequencies there will be one trigger per reference period. At higher reference frequencies there will be one trigger per some multiple of the reference period.

The advantage of this mode over the reference synchronous trigger mode (mode 0)

is that the **ADC 1** input is converted at the same rate as the X- and Y-channel outputs, giving better results when the instrument is set to the ratio display mode.

NOTE: *In the ratio trigger mode, the ADC 2, 3 and 4 inputs are never converted.*

8 “8F” Trigger Mode

In this setting, at reference frequencies below 11 Hz, triggering is synchronous with the reference, but there are eight triggers per reference signal period. On any given trigger, the X and Y channel outputs and one of the four auxiliary **ADC** inputs is converted. Thus any given auxiliary **ADC** input is converted once every four trigger cycles (i.e. twice per reference cycle). The eight X-channel and eight Y-channel output readings are digitally averaged and it is these average values that are displayed and made available for transfer to a computer.

At frequencies above 11 Hz, the trigger mode automatically changes to the asynchronous trigger mode, but 8F triggering is automatically restored if the frequency is subsequently drops to less than 11 Hz.

SAMPLE [n] Main ADC sample rate control

The value of parameter n sets or reads the main ADC sample rate in the range 100 Hz to 200 Hz. To avoid problems with intermodulation with harmonics of the line frequency, the rate is normally factory set to 200 Hz when the instrument is destined for a country with 60 Hz line frequency, and 167 Hz when the line frequency is 50 Hz.

6.4.10 Computer Interfaces (RS232 and GPIB)

RS [n₁ [n₂]] Set or read RS232 interface communications settings

The value of parameter n₁ sets or reads the baud rate of the RS232 interface according to the following table:

n ₁	baud rate (bits per second)
0	75
1	110
2	134.5
3	150
4	300
5	600
6	1200
7	2000
8	2400
9	4800
10	1800
11	9600
12	19200

The lowest six bits in parameter n₂ control the other RS232 parameters according to the following table:

bit number	bit negated	bit asserted
0	7 data bits	8 data bits
1	even parity	odd parity
2	1 parity bit	no parity bit
3	1 stop bit	2 stop bits
4	echo disabled	echo enabled
5	prompt enabled	prompt disabled

GP [n_1 n_2] Set/Read GPIB communications settings

The value of parameter n_1 sets or reads the GPIB address in the range 0 to 31

The value of parameter n_2 sets or reads the GPIB terminator and the test echo settings according to the following table:

n	terminator
0	[CR], test echo disabled
1	[CR], test echo enabled
2	[CR,LF], test echo disabled
3	[CR,LF], test echo enabled
4	No terminator, test echo disabled
5	No terminator, test echo enabled

In all cases the EIO line is asserted with the last byte of a response.

When the test echo is on, every character transmitted or received via the GPIB port is echoed to the RS232 port. This is provided solely as an aid to program development and should not be enabled during normal operation of the instrument.

DD [n] Define delimiter control

The value of parameter n , which can be set to 13 or 32 to 125, determines the ASCII value of the character sent by the lock-in amplifier to separate two numeric values in a two-value response, such as that generated by the MP (magnitude and phase) command.

DRQMODE [n] Data request mode control

The value of parameter n sets or reads the RS232 data request mode status in accordance with the following table:

n	significance
0	Normal operation
1	Data Request Mode active

ST Report status byte

Causes the lock-in amplifier to respond with the status byte, an integer between 0 and 255 which is the decimal equivalent of a binary number with the following bit-significance:

bit	significance
0	command complete
1	invalid command
2	command parameter error
3	reference unlock
4	overload
5	N/A
6	SRQ asserted
7	data available

NOTE: *Unlike the GPIB serial poll, the ST command cannot be used to determine whether an auto-function is in progress (bit 5), since the ST command cannot be issued until the previous command which initiated the auto-function completes.*

The ST command is not normally used in GPIB communications, where the status byte is accessed by performing a serial poll.

N Report overload byte
Causes the lock-in amplifier to respond with the overload byte, an integer between 0 and 255 which is the decimal equivalent of a binary number with the following bit-significance:

bit	significance
0	not used
1	not used
2	not used
3	Y channel output overload
4	X channel output overload
5	PSD overload
6	input overload
7	reference unlock

MSK [n] Set/read service request mask byte
The value of parameter n sets or reads the SRQ mask byte in the range 0 to 255

REMOTE [n] Remote only (front panel lockout) control
The value of parameter n sets or reads the front-panel lock-out control, in accordance with the following table:

n	significance
0	Front panel keys active
1	Front panel keys locked-out, except for the LOCAL key. The instrument can only be controlled via the RS232 or the GPIB interfaces.

6.4.11 Instrument Identification

ID Identification
Causes the lock-in amplifier to respond with the number 5110.

VER	Report firmware version Causes the lock-in amplifier to respond with the firmware version number.
REV	Report firmware revision Causes the lock-in amplifier to respond with the firmware revision number.

6.4.12 Front Panel

D1 [n] Display 1 selection control
The value of parameter n sets or reads the Display 1 selection according to the following table:

n	selection
0	SET REF PH (Reference Phase)
1	SET OSC F (Oscillator Frequency)
2	SET OSC LVL (Oscillator Amplitude)
3	REF F (Display Reference Frequency)
4	DISP (Display output as selected by Display 2 selection control)

D2 [n] Display 2 selection control
The value of parameter n sets or reads the Display 2 selection, and the outputs appearing at the front-panel **CH1** and **CH2** analog outputs according to the following table:

n	selection	CH1 output	CH2 output
0	% FS/X Y	X-channel output	Y-channel output
1	X Y	X-channel output	Y-channel output
2	% FS/ R θ	Magnitude output	Phase Output
3	R θ	Magnitude output	Phase Output
4	OFFSET	X-channel output	Y-channel output
5	NOISE	Noise output	Magnitude output
6	SPEC	Depends on setting of special display modes	

The X-channel and Y-channel outputs are ± 1.000 V corresponding to full-scale, i.e. values of $\pm 100\%$ when in the **% FS** display setting. The magnitude and noise outputs are 0 to $+1.000$ V, corresponding to 0 to 100% of the full-scale sensitivity, and the phase output is ± 1.800 V corresponding to phase angles of $+180.0^\circ$ to -180.0°

NOTE: *There is no command for setting the special display mode number.*

DISP [n] Display update rate control
The value of parameter n, which lies in the range 1 to 10, sets the display update period from 0.1 seconds to 1.0 seconds in 0.1 second increments.

XYDIRECT [n] XY Direct mode control
The output filters used in the model 5110 normally use analog techniques for time-constants of 100 ms or shorter, with digital methods used at longer settings. When the digital filter is used, the **CH1** and **CH2** outputs are driven by the output of a digital to analog converter.

When the XY Direct mode is turned on, the **CH1** and **CH2** outputs are connected directly to the output filters at all time constant settings. Note however that the mode is automatically turned off when any changes are made to relevant instrument controls, such as sensitivity, time constant, slope, etc.

The parameter *n* sets or reads the status of the XY Direct mode in accordance with the following table:

<i>n</i>	selection
0	XY Direct Mode Off. This is the normal setting.
1	XY Direct Mode On

KP Keypress identifier

The KP command reads and resets the status of an internal flag that is set every time a front-panel key is pressed. If the flag is in the set state when the command is issued, the response is 1; if it is in the reset state, then the response is 0. The KP command always leaves the flag in the reset state.

LTS [n] Lights on/off control

The value of parameter *n* controls the front panel LEDs according to the following table:

<i>n</i>	selection
0	All lights off
1	Normal operation
2	All lights on (test mode)

BRIGHT [n] Display LED brightness control

The value of parameter *n* sets or reads the brightness of the front panel seven-segment LEDs over the range 0 to 15.

6.5 Programming Examples

6.5.01 Introduction

This section gives some examples of the commands that need to be sent to the lock-in amplifier for typical experimental situations.

6.5.02 Converting Output Values to Voltages and Currents

Instrument commands which report output values generally do so with a number which represents the output as a proportion of the present full-scale sensitivity, where a response of “+10000” represents 100% of full-scale. If it is desired to know what this represents in terms volts or amps at the input, then the following procedure needs to be adopted:

- 1) Issue a SEN command to determine the present sensitivity setting. Let the response from this command be *senrange*code.

- 2) Determine the full-scale sensitivity range expressed in volts or amps, *senrange*, by the use of a look-up table.
- 3) Issue the command to report the required output. For example, if the X-channel output is required, issue the X command. The response to any of the output commands will be a value between -12000 and +12000, corresponding to $\pm 120.00\%$ of the full-scale sensitivity. Let the reported value be *outputuncal*
- 4) The output can now be expressed in terms of volts or amps as *outputcalib* by the use of the following equations:-

Instrument set to Voltage input mode

$$\text{outputcalib} = \text{outputuncal} \times \text{senrange} \times 10^{-4} \text{ Volts}$$

Instrument set to Current input mode with a 10^6 A/V conversion setting

$$\text{outputcalib} = \text{outputuncal} \times \text{senrange} \times 10^{-10} \text{ Amps}$$

Instrument set to Current input mode with a 10^8 A/V conversion setting

$$\text{outputcalib} = \text{outputuncal} \times \text{senrange} \times 10^{-12} \text{ Amps}$$

NOTE: Remember that the full-scale sensitivity setting is potentially affected by the AS, ASM and SEN commands. Hence, unless it is certain that the setting has not changed, read both the output and the sensitivity setting and follow the above procedure for every output reading.

6.5.03 Basic Signal Recovery

In a typical simple experiment, the computer is used to set the instrument controls and then to record the chosen outputs, perhaps as a function of time. The commands to achieve this would therefore be similar to the following sequence:

IE 2	Set reference source to external analog input mode
LF 0	Turn off line frequency rejection filter
ASM	Auto-Measure
XDB 1	Set output filter slope to 12dB/octave
TC 6	Set time constant to 300ms, since previous ASM may have changed it

Then the outputs could be read as follows:

X	Reads X channel output
Y	Reads Y channel output
MAG	Reads Magnitude
PHA	Reads Phase
FRQ	Reads reference frequency in millihertz

The controlling program would send a new output command each time a new reading were required. Note that a good "rule of thumb" with a 12 dB/octave filter slope is to wait for a period of five time constants after the input signal has changed before recording a new value. Hence in a scanning type experiment, the program should issue the commands to whatever equipment causes the input signal to the lock-in amplifier to change, wait for a period of five time constants, and then record the required output.

6.5.04 Frequency Response Measurement

In this example, the lock in amplifier's internal oscillator output signal is fed via the filter stage under test back to the instrument's signal input. The oscillator frequency is stepped between a lower and an upper frequency and the signal magnitude and phase recorded. The commands to achieve this would therefore be similar to the following sequence:

IE 0	Set reference mode to internal
LF 0	Turn off line frequency rejection filter
OA 1000	Set oscillator amplitude to 1.0 V rms
OF 10000 2	Set oscillator frequency to 100 Hz (starting frequency)
SEN 21	Set sensitivity to 1 V full-scale
TC 4	Set time constant to 30 ms
AQN	Auto-Phase

The frequency sweep would be performed and the outputs recorded by sending the following commands from a FOR...NEXT program loop:

OF n_1 n_2	Set oscillator frequency to new value defined by n_1 and n_2 Software delay of 250ms (5 x 50ms) allowing output to stabilize
MAG	Read Magnitude
PHA	Read Phase
FRQ	Read reference frequency in millihertz. This would nominally be same as the oscillator frequency since the unit is operating in the internal reference mode, but due to oscillator frequency stability considerations, it is more accurate than the value set by the OF n_1 n_2 command.

Repeat these commands until the stop frequency is reached.

Specifications

Appendix A

Measurement Modes

% FS/ X Y	X-channel and Y-channel PSD outputs expressed as a percentage of the present full-scale sensitivity setting
X Y	X-channel and Y-channel PSD outputs expressed directly in terms of voltage or current at input to signal channel
% FS/R θ	Vector magnitude of input signal expressed as a percentage of the present full-scale sensitivity setting and phase angle in degrees
R θ	Vector magnitude of input signal expressed directly in terms of voltage or current at input to signal channel and phase angle in degrees
NOISE	Noise in a bandwidth defined by the output filter time constant and slope controls and centered at the reference frequency expressed as a percentage of the present full-scale sensitivity setting
Harmonic	Fundamental (F) or 2F modes

Displays & Indicators

Two, 3½ -digit seven segment LED displays, analog center-zero panel meter and back-lit LED indicators show the settings of all the main instrument controls and outputs.

Signal Channel

Voltage Inputs	
Modes	A only or Differential (A-B)
Full-scale Sensitivity	100 nV to 1 V rms in a 1-2-5-10 sequence
Impedance	100 M Ω // 25 pF
Maximum Input	$\pm$ 100 V DC; 30 V AC pk-pk without damage, 10 V AC pk-pk without saturation
Voltage Noise	5.5 nV/ $\sqrt{\text{Hz}}$ at 1 kHz typ
CMRR	> 100 dB at 1 kHz degrading by 6 dB/octave
Max CM input	2 V pk-pk
Frequency Response	0.5 Hz to 100 kHz
Grounding	BNC shields can be grounded or floated via 1 k Ω to ground
Current Input	
Mode	10 ⁶ V/A or 10 ⁸ V/A
Full-scale Sensitivity	
10 ⁸ V/A	10 fA to 10 nA in a 1-2-5-10 sequence
10 ⁶ V/A	10 fA to 1 μ A in a 1-2-5-10 sequence

Frequency Response	
10^8 V/A	-3 dB at 330 Hz
10^6 V/A	-3 dB at 60 kHz
Impedance	
10^8 V/A	< 2.5 k Ω at 100 Hz
10^6 V/A	< 250 Ω at 1 kHz
Maximum Input	15 mA continuous, 1 A momentary without damage. 10 μ A AC pk-pk without saturation on 10^6 V/A; 100 nA AC pk-pk without saturation on 10^8 V/A
Noise	
10^8 V/A	13 fA/ $\sqrt{\text{Hz}}$ at 500 Hz
10^6 V/A	130 fA/ $\sqrt{\text{Hz}}$ at 1 kHz
Grounding	BNC shield can be grounded or floated via 1 k Ω to ground
Line Notch Filter	> 34dB attenuation @ $\pm 1\%$ of 50 or 60 Hz and/or 100 or 120 Hz
Dynamic Reserve	105 dB max
Gain Accuracy	
Flat Mode	$\pm 1\%$ typical
Bandpass Mode	$\pm 2\%$ typical
Gain Stability	200ppm/ $^{\circ}\text{C}$ typical

Reference Channel

TTL Input (rear panel)	
Frequency Range	0.5 Hz to 100 kHz
Analog Input (front panel)	
Frequency Range	0.5 Hz to 100 kHz
Impedance	1 M Ω // 30 pF
Level	100 mV to 5 V rms sinewave or TTL
Maximum input voltage	5.0 V rms
Phase	
Set Resolution	0.1 $^{\circ}$, 0.025 $^{\circ}$ or 0.005 $^{\circ}$ increments
Accuracy	$\pm 1^{\circ}$ typical
Noise	0.005 $^{\circ}$ rms at 100 ms TC, 12 dB/octave
Orthogonality	
Above 5Hz	90 $^{\circ}$ $\pm 0.5^{\circ}$
0.5Hz - 5Hz	90 $^{\circ}$ $\pm 5^{\circ}$ max
Drift (Flat Mode)	< 0.05 $^{\circ}$ / $^{\circ}\text{C}$
Lock Acquisition Time	1 second +2 cycles max

Demodulators

Description	Switching type demodulators operating in either square wave or Walsh function modes.
Output Zero Stability	
High Dynamic Reserve	500 ppm/°C
Normal	50 ppm/°C
High Stability	5 ppm/°C
Harmonic Rejection	
Low-Pass	>80dB at 1 kHz
Bandpass	>60dB at 1 kHz
Time Constants	
Outputs	Min ($\leq 20 \mu\text{s}$) and 1 ms to 300 s in a 1-3-10 sequence
Roll-off	6 and 12 dB/octave
Offset	Auto and Manual on X and/or Y: $\pm 100\%$ FS

Oscillator

Frequency	
Range	0.5 Hz to 100 kHz
Setting Resolution	better than 1%
Absolute Accuracy	$\pm 2\%$
Distortion (THD)	0.5%
Amplitude	
Range	
Front panel	1 mV to 1.999 V
Computer Control	1 mV to 2.000 V and 5.000 V
Setting Resolution	
1 mV to 500 mV	1 mV
501 mV to 2 V	8 mV
Output	
Impedance	450 Ω

Auxiliary Inputs

ADC 1 - CH4	
Maximum Input	$\pm 12\text{ V}$
Resolution	1 mV
Input Impedance	1 M Ω // 30 pF
Sample Rate	
CH1 only	200 Hz max.
CH1 - CH4	50 Hz max.

Trigger Mode	Internal or External
Trigger input	TTL compatible
Outputs	
CH1, CH2 Analog Outputs	
Function	X, Y, R, θ , Noise and Ratio.
Amplitude	± 1.2 V (± 1.0 V = $\pm$ full scale)
Impedance	1 k Ω
Signal Monitor	
Amplitude	± 1.0 V max
Impedance	1 k Ω
Aux D/A Outputs	
Maximum Output	± 12 V
Resolution	1 mV
Output Impedance	< 150 Ω
8-bit Digital Output	
	8 TTL-compatible lines that can be independently set high or low to activate external equipment
Reference Output	
Waveform	0 to 5 V square wave
Impedance	TTL compatible
Power - Low Voltage	
	± 15 V at 50 mA rear panel DIN connector for powering SIGNAL RECOVERY preamplifiers

Interfaces

RS232 and GPIB (IEEE-488). All settings that can be adjusted from the front-panel, except A, A-B, 10^6 V/A, 10^8 V/A and Float/Ground, can be set and read, and all instrument outputs can be read, via these interfaces.

General

Power Requirements	
Voltage	110/120/220/240 VAC
Frequency	50/60 Hz
Dimensions	
Width	440 mm (17.25")
Depth	500 mm (19.5 ")

Height	
With feet	105 mm (4.1 ")
Without feet	89 mm (3.5 ")
Weight	
	7.6 kg (16.8 lbs)

All specifications subject to change without notification

B1 RS232 Connector Pinout

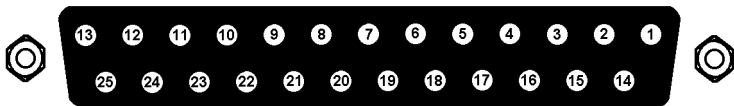


Figure B-1, RS232 Connector (Female)

Pin	Function	Description
1	Earth Ground	Ties the chassis of the model 5110 to that of the computer
2	Transmit Data	The 5110 transmits data on this line
3	Receive Data	The 5110 receives data on this line
4	Request to Send	Constantly asserted
7	Logic Ground	Data signals are referenced with respect to the voltage at this pin
20	Data Terminal Ready	Constantly asserted

All other pins are not connected

B2 Preamplifier Power Connector Pinout

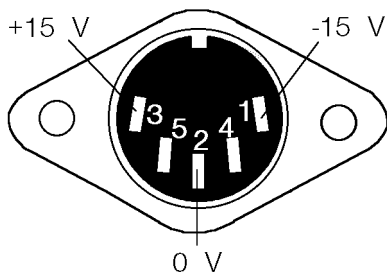


Figure B-2, Preamplifier Power Connector

Pin	Function
1	-15 V
2	Ground
3	+15 V

Pins 4 and 5 are not connected. Shell is shield ground.

B3 Digital Output Port Connector

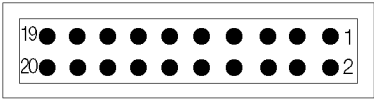


Figure B-3, Digital Output Port Connector

8-bit TTL-compatible output set from the front panel or via the computer interfaces. Each line can drive 3 LSTTL loads. The connector will mate with a 20-pin IDC header plug (not supplied). The pinout is as follows:-

Pin	Function
1	Ground
2	Ground
3	D0
4	Ground
5	D1
6	Ground
7	D2
8	Ground
9	D3
10	Ground
11	D4
12	Ground
13	D5
14	Ground
15	D6
16	Ground
17	D7
18	Ground
19	+5 V
20	+5 V

D0 = Least Significant Bit
D7 = Most Significant Bit

Demonstration Programs

Appendix C

C1 Simple Terminal Emulator

This is a short terminal emulator with minimal facilities, which will run on a PC-compatible computer in a Microsoft GWBASIC or QuickBASIC environment, or can be compiled with a suitable compiler.

```
10 'MINITERM 9-Feb-96
20 CLS : PRINT "Lockin RS232 parameters must be set to 9600 baud, 7 DATA bits, 1 stop
bit and even parity"
30 PRINT "Hit <ESC> key to exit"
40 OPEN "COM1:9600,E,7,1,CS,DS" FOR RANDOM AS #1
50 '.....
60 ON ERROR GOTO 180
70 '.....
100 WHILE (1)
110     B$ = INKEY$
120     IF B$ = CHR$(27) THEN CLOSE #1: ON ERROR GOTO 0: END
130     IF B$ <> "" THEN PRINT #1, B$;
140     LL% = LOC(1)
150     IF LL% > 0 THEN A$ = INPUT$(LL%, #1): PRINT A$;
160 WEND
170 '.....
180 PRINT "ERROR NO."; ERR: RESUME
```

C2 RS232 Control Program with Handshakes

RSCOM2.BAS is a user interface program which illustrates the principles of the echo handshake. The program will run on a PC-compatible computer either in a Microsoft GWBASIC or QuickBASIC environment, or in compiled form.

The subroutines in RSCOM2 are recommended for incorporation in the user's own programs.

```
10 'RSCOM2 9-Feb-96
20 CLS : PRINT "Lockin RS232 parameters must be set to 9600 baud, 7 data bits, 1 stop
bit, even parity"
30 OPEN "COM1:9600,E,7,1,CS,DS" FOR RANDOM AS #1
40 CR$ = CHR$(13) ' carriage return
50 '
60 '...main loop.....
70 WHILE 1 ' infinite loop
80     INPUT "command (00 to exit) "; B$ ' no commas are allowed in B$
90     IF B$ = "00" THEN END
100    B$ = B$ + CR$ ' append a carriage return
110    GOSUB 180 ' output the command B$
```

```

120     GOSUB 310: PRINT Z$;           ' read and display response
130     IF A$ = "?" THEN GOSUB 410: GOSUB 470 ' if "?" prompt fetch STATUS%
140                                           ' and display message
150 WEND                               ' return to start of loop
160 '
170 '
180 '...output the string B$.....
190 ON ERROR GOTO 510                 ' enable error trapping
200 IF LOC(1) > 0 THEN A$ = INPUT$(LOC(1), #1) ' clear input buffer
210 ON ERROR GOTO 0                   ' disable error trapping
220 FOR J1% = 1 TO LEN(B$)           ' LEN(B$) is number of bytes
230     C$ = MID$(B$, J1%, 1): PRINT #1, C$; ' send byte
240     WHILE LOC(1) = 0: WEND         ' wait for byte in input buffer
250     A$ = INPUT$(1, #1)            ' read input buffer
260     IF A$ <> C$ THEN PRINT "handshake error" ' input byte should be echo
270 NEXT J1%                          ' next byte to be sent or
280 RETURN                           ' return if no more bytes
290 '
300 '
310 '....read response.....
320 A$ = "": Z$ = ""
330 WHILE (A$ <> "*" AND A$ <> "?")    ' read until prompt received
340     Z$ = Z$ + A$                  ' append next byte to string
350     WHILE LOC(1) = 0: WEND         ' wait for byte in input buffer
360     A$ = INPUT$(1, #1)            ' read byte from buffer
370 WEND                             ' next byte to be read
380 RETURN                           ' return if it is a prompt
390 '
400 '
410 '....fetch status byte.....
420 B$ = "ST" + CR$                  ' "ST" is the status command
430 GOSUB 180                         ' output the command
440 GOSUB 310                         ' read response into Z$
450 STATUS% = VAL(Z$)                ' convert to integer
460 RETURN
470 '....instrument error message.....
480 PRINT "Error prompt, status byte = "; STATUS% ' bits are defined in manual
490 PRINT
500 RETURN
510 '....I/O error routine.....
520 RESUME

```

C3 GPIB User Interface Program

GPCOM.BAS is a user interface program which illustrates the principles of the use of the serial poll status byte to coordinate the command and data transfer.

The program runs under Microsoft GWBASIC or QuickBASIC on a PC-compatible computer fitted with a National Instruments IEEE488 interface card and the GPIB.COM software installed in the CONFIG.SYS file. The program BIB.M, and the first three lines of GPCOM, are supplied by the card manufacturer and must be the correct version for the particular version of the interface card in use. The interface card may be set up, using the program IBCONF.EXE, to set EOI with the last byte of Write in which case no terminator is required. (Read operations are automatically terminated on EOI which is always sent by the lock-in amplifier). Normally, the options called 'high-speed timing', 'interrupt jumper setting', and 'DMA channel' should all be disabled.

The principles of using the Serial Poll Status Byte to control data transfer, as implemented in the main loop of GPCOM, are recommended for incorporation in the user's own programs.

```

10 'GPCOM 9-Feb-96
20 '....the following three lines and BIB.M are supplied by the.....
30 '....manufacturer of the GPIB card, must be correct version.....
40 CLEAR , 60000!: IBINIT1 = 60000!: IBINIT2 = IBINIT1 + 3: BLOAD "BIB.M", IBINIT1
50 CALL IBINIT1(IBFIND, IBTRG, IBCLR, IBPCT, IBSIC, IBLOC, IBPPC, IBBNA, IBONL, IBRSC,
IBSRE, IBRSV, IBPAD, IBSAD, IBIST, IBDMA, IBEOS, IBTMO, IBEOT, IBRDF, IBWRTF, IBTRAP)
60 CALL IBINIT2(IBGTS, IBCAC, IBWAIT, IBPOKE, IBWRT, IBWRTA, IBCMD, IBCMDA, IBRD, IBRDA,
IBSTOP, IBRPP, IBRSP, IBDIAG, IBXTRC, IBRDI, IBWRTI, IBRDIA, IBWRTIA, IBSTA%, IBERR%,
IBCNT%)
70 '.....
80 CLS : PRINT "DEVICE MUST BE SET TO CR TERMINATOR"
90 '....assign access code to interface board.....
100 BDNAM$ = "GPIB0"
110 CALL IBFIND(BDNAM$, GPIB0%)
120 IF GPIB0% < 0 THEN PRINT "board assignment error": END
130 '....send INTERFACE CLEAR.....
140 CALL IBSIC(GPIB0%)
150 '....set bus address, assign access code to device.....
160 SUCCESS% = 0
170 WHILE SUCCESS% = 0
180     INPUT "BUS ADDRESS "; A%
190     DEVNAME$ = "DEV" + RIGHT$(STR$(A%), LEN(STR$(A%)) - 1)
200     CALL IBFIND(DEVNAME$, DEV%)          ' assign access code
210     IF DEV% < 0 THEN PRINT "device assignment error": END
220     A$ = CHR$(13): GOSUB 480              ' test: write <CR> to bus
230     IF IBSTA% > 0 THEN SUCCESS% = 1
240     IF (IBSTA% < 0 AND IBERR% = 2) THEN BEEP: PRINT "NO DEVICE AT THAT ADDRESS ";
250 WEND
260 '....send SELECTED DEVICE CLEAR.....
270 CALL IBCLR(DEV%)

```

```

280 '....set timeout to 1 second.....
290 V% = 11: CALL IBTMO(DEV%, V%)
300 '....set status print flag.....
310 INPUT "Display status byte y/n "; R$
320 IF R$ = "Y" OR R$ = "y" THEN DS% = 1 ELSE DS% = 0
330 '....main loop.....
340 WHILE 1                                ' infinite loop
350     INPUT "command (00 to exit) "; A$
360     IF A$ = "00" THEN END
370     A$ = A$ ' CHR$(13)                    ' terminator is <CR>
380     GOSUB 480                            ' write A$ to bus
390     S% = 0                              ' initialize S%
400     WHILE (S% AND 1) = 0                ' while command not complete
410         GOSUB 530                      ' serial poll, returns S%
420         IF DS% THEN PRINT "S%= "; S%
430         IF (S% AND 128) THEN GOSUB 500: PRINT B$ ' read bus into B$ and print
440     WEND
445     IF (S% AND 4) THEN PRINT "parameter error"
450     IF (S% AND 2) THEN PRINT "invalid command"
460 WEND
470 '....end of main loop.....
480 '....write string to bus.....
490 CALL IBWRT(DEV%, A$): RETURN
500 '....read string from bus.....
510 B$ = SPACE$(32)                        ' B$ is buffer
520 CALL IBRD(DEV%, B$): RETURN
530 '.....serial poll.....
540 CALL IBRSP(DEV%, S%): RETURN

```

Cable Diagrams

D1 RS232 Cable Diagrams

Users who choose to use the RS232 interface to connect the model 5110 lock-in amplifier to a standard serial port on a computer will need to use one of two types of cable. The only difference between them is the number of pins used on the connector which goes to the computer. One has 9 pins and the other 25; both are null-modem (also called modem eliminator) cables in that some of the pins are cross-connected.

Users with reasonable practical skills can easily assemble the required cables from parts which are widely available through computer stores and electronics components suppliers. The required interconnections are given in figures D-1 and D-2

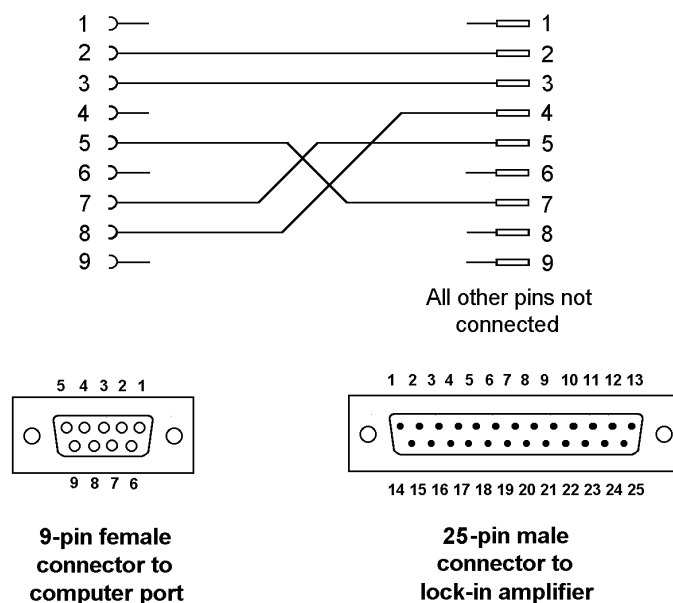


Figure D-1, Interconnecting RS232 Cable Wiring Diagram

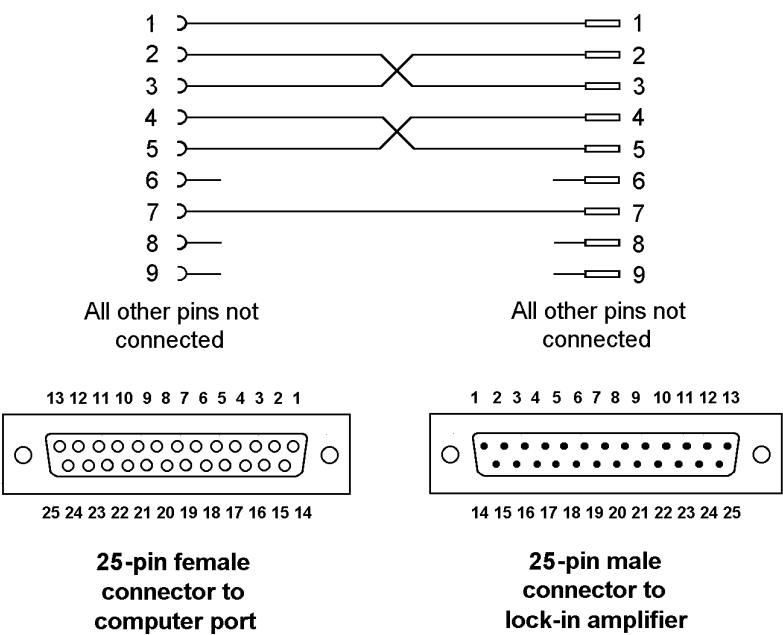


Figure D-2, Interconnecting RS232 Cable Wiring Diagram

Alphabetical Listing of Commands

Appendix E

- ADC n** Read auxiliary analog to digital inputs
Reads the voltage appearing at the rear panel **ADC 1** (n = 1) or , **2** (n = 2), **3** (n = 3) or **4** (n = 4) inputs.
- The response for ADC n is an integer in the range -12000 to +12000, corresponding to voltages from -12.000 V to +12.000 V.
- AQN** Auto-Phase (auto quadrature null)
The instrument adjusts the reference phase to maximize the X channel output and minimize the Y channel output signals.
- AS** Perform an Auto-Sensitivity operation
The instrument adjusts its full-scale sensitivity so that the magnitude output lies between 20 % and 90 % of full-scale.
- ASM** Perform an Auto-Measure operation
The instrument adjusts its full-scale sensitivity so that the magnitude output lies between 20 % and 90 % of full-scale, and then performs an auto-phase operation to maximize the X-channel output and minimize the Y-channel output.
- ATC [n]** Filter frequency tuning mode control
The value of parameter n sets or reads the filter frequency tuning mode according to the following table:
- | | |
|---|---|
| n | filter frequency tuning mode |
| 0 | manual (frequency set by front-panel keys or by the FF command) |
| 1 | automatic (filter frequency set to reference channel frequency) |
- ATS** Auto-tune filter frequency
The ATS command causes the filter frequency to be tuned to the reference frequency, by setting the filter track mode on and then off.
- AXO** Auto-Offset
The X and Y channel output offsets are turned on and set to levels giving zero X and Y channel outputs. Any changes in the input signal then appear as changes about zero in the outputs.
- BRIGHT [n]** Display LED brightness control
The value of parameter n sets or reads the brightness of the front panel seven-segment LEDs over the range 0 to 15.
- BYTE [n]** Digital output port control
The value of n, in the range 0 to 255, determines the bits to be output on the rear panel digital output port. When n = 0, all outputs are low, and when n = 255, all are high.

D1 [n] Display 1 selection control
The value of parameter n sets or reads the Display 1 selection according to the following table:

n	selection
0	SET REF PH (Reference Phase)
1	SET OSC F (Oscillator Frequency)
2	SET OSC LVL (Oscillator Amplitude)
3	REF F (Display Reference Frequency)
4	DISP (Display output as selected by Display 2 selection control)

D2 [n] Display 2 selection control
The value of parameter n sets or reads the Display 2 selection, and the outputs appearing at the front-panel **CH1** and **CH2** analog outputs according to the following table:

n	selection	CH1 output	CH2 output
0	% FS/X Y	X-channel output	Y-channel output
1	X Y	X-channel output	Y-channel output
2	% FS/ R θ	Magnitude output	Phase Output
3	R θ	Magnitude output	Phase Output
4	OFFSET	X-channel output	Y-channel output
5	NOISE	Noise output	Magnitude output
6	SPEC	Depends on setting of special display modes	

The X-channel and Y-channel outputs are ± 1.000 V corresponding to full-scale, i.e. values of $\pm 100\%$ when in the **% FS** display setting. The magnitude and noise outputs are 0 to $+1.000$ V, corresponding to 0 to 100% of the full-scale sensitivity, and the phase output is ± 1.800 V corresponding to phase angles of $+180.0^\circ$ to -180.0°

NOTE: *There is no command for setting the special display mode number.*

DAC1 [n] Auxiliary DAC 1 output control
The value of parameter n sets or reads the voltage appearing at the rear panel **DAC 1** output, in the range -12000 to +12000, corresponding to voltages from -12.000 V to +12.000 V.

DAC2 [n] Auxiliary DAC 2 output control
The value of parameter n sets or reads the voltage appearing at the rear panel **DAC 2** output, in the range -12000 to +12000, corresponding to voltages from -12.000 V to +12.000 V.

DD [n] Define delimiter control
The value of parameter n, which can be set to 13 or 32 to 125, determines the ASCII value of the character sent by the lock-in amplifier to separate two numeric values in a two-value response, such as that generated by the MP (magnitude and phase) command.

- DISP [n]** Display update rate control
The value of parameter n, which lies in the range 1 to 10, sets the display update period from 0.1 seconds to 1.0 seconds in 0.1 second increments.
- DR [n]** Dynamic reserve control
The value of parameter n sets or reads the instrument's dynamic reserve, according to the following table:
- | | |
|---|--|
| n | selection |
| 0 | HI STAB (high stability setting) |
| 1 | NORM (normal setting) |
| 2 | HI RES (high dynamic reserve setting) |
- DRQMODE [n]** Data request mode control
The value of parameter n sets or reads the RS232 data request mode status in accordance with the following table:
- | | |
|---|--------------------------|
| n | significance |
| 0 | Normal operation |
| 1 | Data Request Mode active |
- EX [n]** Output expansion control
The value of parameter n sets or reads the X-channel output expansion mode in accordance with the following table:
- | | |
|---|--|
| n | expand mode |
| 0 | Off |
| 1 | Expand X-channel output by $\times 10$. |
- F2F [n]** Reference harmonic mode control
The value of parameter n sets or reads the reference frequency mode, according to the following table:
- | | |
|---|---|
| n | selection |
| 0 | Signal detection occurs at the applied reference frequency (internal or external) |
| 1 | Signal detection occurs at twice the applied reference frequency (internal or external) |
- FF [n₁, n₂]** Filter frequency control
The value of parameter n₂ sets or reads the filter frequency band according to the following table:
- | | |
|----------------|-------------------|
| n ₂ | frequency band |
| 0 | 0.5 Hz to 12 Hz |
| 1 | 10 Hz to 120 Hz |
| 2 | 100 Hz to 1200 Hz |
| 3 | 1 kHz to 12 kHz |
| 4 | 10 kHz to 105 kHz |

The value of parameter n₁ sets or reads the frequency within each band in the range

1000 to 12000, except in the bottom band ($n_2 = 0$) where the range is 500 to 12000 and in the top band ($n_2 = 4$) where the range is 1000 to 10500

Hence, for example, to set the frequency to 1 kHz, send the commands FF 1000 3 or FF 10000 2

FLT [n] Main filter operating mode
The value of parameter n sets or reads the signal-channel filter mode according to the following table:

n	mode
0	FLAT (no filter)
1	LP (low-pass filter)
2	BP (band-pass filter)

FRQ Reference frequency meter
The FRQ command causes the lock-in amplifier to respond with 0 if the reference channel is unlocked, or with the measured reference frequency expressed in millihertz if it is locked.

GP [n_1 n_2] Set/Read GPIB communications settings
The value of parameter n_1 sets or reads the GPIB address in the range 0 to 31

The value of parameter n_2 sets or reads the GPIB terminator and the test echo settings according to the following table:

n	terminator
0	[CR], test echo disabled
1	[CR], test echo enabled
2	[CR,LF], test echo disabled
3	[CR,LF], test echo enabled
4	No terminator, test echo disabled
5	No terminator, test echo enabled

In all cases the EIO line is asserted with the last byte of a response.

When the test echo is on, every character transmitted or received via the GPIB port is echoed to the RS232 port. This is provided solely as an aid to program development and should not be enabled during normal operation of the instrument.

ID Identification
Causes the lock-in amplifier to respond with the number 5110.

IE [n] Reference channel source control (Internal/External)
The value of parameter n sets or reads the reference input mode according to the following table:

- n selection
- 0 INT (Internal)
- 1 EXT TTL (External rear-panel **TTL REF IN** input)
- 2 EXT Analog (External front panel **REF IN** input)

KP Keypress identifier

The KP command reads and resets the status of an internal flag that is set every time a front-panel key is pressed. If the flag is in the set state when the command is issued, the response is 1; if it is in the reset state, then the response is 0. The KP command always leaves the flag in the reset state.

LF [n] Signal channel line frequency rejection filter control

The value of parameter n sets or reads the mode of the line frequency notch filter according to the following table:

- n selection
- 0 Off
- 1 Enable 50 or 60 Hz notch filter
- 2 Enable 100 or 120 Hz notch filter
- 3 Enable both filters

LINE [n] Line frequency rejection filter center frequency control

The value of parameter n sets or reads the center frequencies of the line frequency notch filter according to the following table:

- n selection
- 0 50 and/or 100 Hz
- 1 60 and/or 120 Hz

LOCK Lock reference channel

Causes a reference lock operation, updating all frequency-dependent calibration parameters.

LTS [n] Lights on/off control

The value of parameter n controls the front panel LEDs according to the following table:

- n selection
- 0 All lights off
- 1 Normal operation
- 2 All lights on (test mode)

MAG Magnitude

The MAG command causes the lock-in amplifier to respond with the magnitude output in the range 0 to +12000, full-scale being +10000

MP Magnitude, phase

Equivalent to the compound command MAG;PHA

MSK [n] Set/read service request mask byte
The value of parameter n sets or reads the SRQ mask byte in the range 0 to 255

N Report overload byte
Causes the lock-in amplifier to respond with the overload byte, an integer between 0 and 255 which is the decimal equivalent of a binary number with the following bit-significance:

bit	significance
0	not used
1	not used
2	not used
3	Y channel output overload
4	X channel output overload
5	PSD overload
6	input overload
7	reference unlock

NN Noise output
The NN command causes the lock-in amplifier to respond with the noise, expressed as a percentage of the present full scale sensitivity, with 10000 corresponding to 100%. The noise is measured in a bandwidth defined by the output filter time constant and slope, centered about the reference frequency.

OA [n] Oscillator amplitude control
The value of parameter n sets or reads the oscillator amplitude. The range of n is 0 to 2000 representing 0 to 2 V rms, but the value 5000 may also be used, which corresponds to an output amplitude of 5.000 V rms.

OF [n₁ n₂] Oscillator frequency control
The value of parameter n₂ sets or reads the oscillator frequency band according to the following table:

n ₂	frequency band
0	0.5 Hz to 2 Hz
1	1.8 Hz to 20 Hz
2	18 Hz to 200 Hz
3	180 Hz to 2 kHz
4	1.8 kHz to 20 kHz
5	18 kHz to 105 kHz

The value of parameter n₁ sets or reads the frequency within each band in the range 1800 to 20000, except for the bottom band (n₂ = 0) where the range is 5000 to 20000 and the top band (n₂ = 5) where the range is 1800 to 10500.

Hence, for example, to set the frequency to 1 kHz, send the command OF 10000 3.

P [n₁, n₂] Reference phase control
The parameters n₁ and n₂ set or read the reference channel phase shifter, as follows:-

Parameter n_1 sets or reads the phase quadrant according to the following table:

n_1	quadrant
0	0° to 90°
1	90° to 180°
2	180° to 270°
3	270° to 360°

Parameter n_2 sets or reads the phase within each quadrant to any value between 0° and 100° in millidegrees, with a minimum resolution of 5 m°, i.e. n_2 can range from 0 to 100000.

PHA Signal phase

The PHA command causes the lock-in amplifier to respond with the signal phase in millidegrees, in the range ± 180000 , corresponding to $\pm 180^\circ$.

REMOTE [n] Remote only (front panel lockout) control

The value of parameter n sets or reads the front-panel lock-out control, in accordance with the following table:

n	significance
0	Front panel keys active
1	Front panel keys locked-out, except for the LOCAL key. The instrument can only be controlled via the RS232 or the GPIB interfaces.

REV Report firmware revision

Causes the lock-in amplifier to respond with the firmware revision number.

RS [n_1 [n_2]] Set or read RS232 interface communications settings

The value of parameter n_1 sets or reads the baud rate of the RS232 interface according to the following table:

n_1	baud rate (bits per second)
0	75
1	110
2	134.5
3	150
4	300
5	600
6	1200
7	2000
8	2400
9	4800
10	1800
11	9600
12	19200

The lowest six bits in parameter n_2 control the other RS232 parameters according to the following table:

bit number	bit negated	bit asserted
0	7 data bits	8 data bits
1	even parity	odd parity
2	1 parity bit	no parity bit
3	1 stop bit	2 stop bits
4	echo disabled	echo enabled
5	prompt enabled	prompt disabled

RT Ratio output

The RT command reports a number equivalent to $(1000 \times X)/ADC1$ where X is the value that would be returned by the X command and ADC1 is the value that would be returned by the ADC1 command.

SAMPLE [n] Main ADC sample rate control

The value of parameter n sets or reads the main ADC sample rate in the range 100 Hz to 200 Hz. To avoid problems with intermodulation with harmonics of the line frequency, the rate is normally factory set to 200 Hz when the instrument is destined for a country with 60 Hz line frequency, and 167 Hz when the line frequency is 50 Hz.

SEN [n] Full-scale sensitivity control

The value of parameter n sets or reads the full-scale sensitivity according to the following table:

n	full-scale sensitivity
0	100 nV
1	200 nV
2	500 nV
3	1 μ V
4	2 μ V
5	5 μ V
6	10 μ V
7	20 μ V
8	50 μ V
9	100 μ V
10	200 μ V
11	500 μ V
12	1 mV
13	2 mV
14	5 mV
15	10 mV
16	20 mV
17	50 mV
18	100 mV
19	200 mV
20	500 mV
21	1 V

ST Report status byte

Causes the lock-in amplifier to respond with the status byte, an integer between 0 and 255 which is the decimal equivalent of a binary number with the following bit-

significance:

bit	significance
0	command complete
1	invalid command
2	command parameter error
3	reference unlock
4	overload
bit	significance
5	N/A
6	SRQ asserted
7	data available

NOTE: Unlike the GPIB serial poll, the ST command cannot be used to determine whether an auto-function is in progress (bit 5), since the ST command cannot be issued until the previous command which initiated the auto-function completes.

The ST command is not normally used in GPIB communications, where the status byte is accessed by performing a serial poll.

TRIG [n] ADC trigger mode control

The value of parameter n sets or reads the trigger mode of the ADC converter used for converting both the X-channel and Y-channel outputs and the auxiliary ADC inputs according to the following table:

n trigger mode

0 Reference Trigger Mode

In this setting, the ADC is triggered synchronously with the reference frequency. The maximum conversion rate is nominally once every 5 ms (200 Hz), so that at low reference frequencies there will be one trigger per reference period. At higher reference frequencies there will be one trigger per some multiple of the reference period.

Each trigger results in the conversion of X- and Y-channel outputs and one of the four auxiliary ADC inputs. Thus any given auxiliary ADC input is converted once every four trigger cycles.

1 External Trigger Mode

In this setting the ADC is triggered once on each positive going TTL transition detected at rear-panel **ADC EXT TRIG IN** connector. If an applied trigger occurs before the converter has finished processing the previous trigger, it is ignored. On any given trigger, the X- and Y-channel outputs and all four of the auxiliary ADC inputs are converted.

2 Asynchronous Trigger Mode

In this setting the ADC is triggered asynchronously at the preset system sample rate, which is adjustable over the range 100 Hz to 200 Hz by means of a front panel control or the SAMPLE computer command. On any given trigger, the X- and Y-channel outputs and one of the four auxiliary ADC inputs are converted. Thus any given auxiliary ADC input is converted once every four trigger cycles

(i.e. at a quarter of the system sample rate).

4 Ratio Trigger Mode

In this setting, triggering is synchronous with the reference signal as for mode 0, but only the X- and Y- channel outputs and the **ADC AUX INPUT CH1** are converted. The maximum conversion rate is nominally once every 5 ms (200 Hz), so that at low reference frequencies there will be one trigger per reference period. At higher reference frequencies there will be one trigger per some multiple of the reference period.

The advantage of this mode over the reference synchronous trigger mode (mode 0) is that the **ADC AUX INPUT CH1** input is converted at the same rate as the X- and Y-channel outputs, giving better results when the instrument is set to the ratio display mode.

NOTE: In the ratio trigger mode, the ADC 2, 3 and 4 inputs are never converted.

8 “8F” Trigger Mode

In this setting, at reference frequencies below 11 Hz, triggering is synchronous with the reference, but there are eight triggers per reference signal period. On any given trigger, the X and Y channel outputs and one of the four auxiliary **ADC** inputs is converted. Thus any given auxiliary **ADC** input is converted once every four trigger cycles (i.e. twice per reference cycle). The eight X-channel and eight Y-channel output readings are digitally averaged and it is these average values that are displayed and made available for transfer to a computer.

At frequencies above 11 Hz, the trigger mode automatically changes to the asynchronous trigger mode, but 8F triggering is automatically restored if the frequency is subsequently drops to less than 11 Hz.

VER Report firmware version

Causes the lock-in amplifier to respond with the firmware version number.

X X channel output

The X command causes the lock-in amplifier to respond with the X-channel output in the range ± 12000 , full-scale being ± 10000

XDB [n] Output low-pass filter slope (roll-off) control

The value of parameter n sets or reads the slope of the output filters according to the following table:

n	slope
0	6dB/octave
1	12dB/octave

XOF [n₁ [n₂]] X channel output offset control

The value of parameter n₁ sets or reads the status and level of the X offset facility according to the following table:

n_1 selection
 0 Disables X offset facility
 1 Enables X offset facility

The range of n_2 is ± 1000 corresponding to $\pm 100\%$ of the full-scale output.

XTC [n] or

TC [n] Filter time constant control

The value of parameter n sets or reads the time constant of the output according to the following table:

n	time constant
0	MIN
1	1 ms
2	3 ms
3	10 ms
4	30 ms
5	100 ms
6	300 ms
7	1 s
8	3 s
9	10 s
10	30 s
11	100 s
12	300 s

XY X, Y channel outputs

Equivalent to the compound command X;Y

XYDIRECT [n] XY Direct mode control

The output filters used in the model 5110 normally use analog techniques for time-constants of 100 ms or shorter, with digital methods used at longer settings. When the digital filter is used, the **CH1** and **CH2** outputs are driven by the output of a digital to analog converter.

When the XY Direct mode is turned on, the **CH1** and **CH2** outputs are connected directly to the output filters at all time constant settings. Note however that the mode is automatically turned off when any changes are made to relevant instrument controls, such as sensitivity, time constant, slope, etc.

The parameter n sets or reads the status of the XY Direct mode in accordance with the following table:

n	selection
0	XY Direct Mode Off. This is the normal setting.
1	XY Direct Mode On

Y Y channel output

The Y command causes the lock-in amplifier to respond with the Y-channel output in the range ± 12000 , full-scale being ± 10000

YOF [n_1 [n_2]] Y channel output offset control

The value of parameter n_1 sets or reads the status and level of the Y offset facility according to the following table:

n_1	selection
0	Disables Y offset facility
1	Enables Y offset facility

The range of n_2 is ± 1000 corresponding to ± 100 % of the full-scale output.

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|---|---|
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